600 Jahre Universität Rostock

Traditio et Innovatio

Master's Thesis

End-to-End AI Pipeline for Cow Re-Identification in Dairy Farms

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Enrolment number:	223100134
Work period:	01. April 2026 – 19. August 2026
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Abstract

Reliable identification of individual cows is important for linking visual observations to the correct animal in dairy farm monitoring. This is especially useful for applications such as health monitoring, lameness detection, feeding analysis, and welfare assessment. This thesis develops and evaluates an end-to-end computer-vision pipeline for individual cow re-identification in dairy farm environments. The system was evaluated using two farm-specific scenarios, referred to as Farm A and Farm B, with different recording and occlusion conditions.

ResNet-50, MegaDescriptor, and DINOv3 were compared for image-level Cow Re-ID, and DINOv3 produced the strongest overall results. The final pipeline combines cow localisation and segmentation, DINOv3 embedding extraction, gallery-based cosine-similarity matching, BoT-SORT tracking, and track-level identity stabilisation. Each enrolled cow is represented by a prototype embedding created from multiple gallery images. For Farm B, a custom YOLO11m segmentation model was trained using SAM3-generated pseudo-labels to better handle strong vertical metal-bar occlusion.

For Farm A, the 15 cows used in the inference gallery were not included during DINOv3 training, validation, or testing. Across 30 daytime and night-time target videos, the system correctly identified 23 cases, giving an overall identification rate of 76.67%. For Farm B, the final DINOv3 and custom YOLO11m configuration recovered 87 of 122 RFID-confirmed gallery identities. Under the RFID-supported temporal evaluation, it achieved 93.55% Precision, 71.31% Recall, an F1-score of 80.93%, and 78.53% Accuracy.

The results show that gallery-based Cow Re-ID can provide useful individual identification from farm videos. They also show that practical video Re-ID can be influenced by the embedding model, segmentation quality, visible coat information, recording conditions, tracking, and stable identity evidence across multiple frames.

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1 Introduction

Individual cow identification is important for modern dairy farm monitoring. Many practical applications, such as health monitoring, lameness detection, feeding analysis, and welfare assessment, become more useful when the observed information can be connected to the correct individual cow [1]. Traditional identification methods such as ear tags and RFID are widely used, but they have limitations for continuous camera-based monitoring [2]. For example, RFID requires dedicated readers, while camera-based reading of ear tags depends on the tag being visible in the image.

Vision-based cow re-identification, or Cow Re-ID, provides an alternative way to identify individual cows directly from images and videos. Unlike cow detection, which determines where a cow is located, Cow Re-ID determines which individual cow is being observed. For cows with distinctive coat patterns, visual appearance can provide useful identity information [3, 4].

However, Cow Re-ID in real farm videos is challenging. Lighting conditions can change, cows may be partially occluded, only part of the body may be visible, and background structures can interfere with the extracted visual features. In video data, identification results can also change between frames because of movement, segmentation errors, or temporary tracking problems.

Therefore, a practical Cow Re-ID system requires several components to work together. The cow must first be located and separated from the background. Visual identity features must then be extracted and compared with reference identities. In video, the cow should also be tracked across frames so that identity evidence can be accumulated before a stable cow ID is displayed.

Previous research has demonstrated visual cattle identification in image- and video-based settings [3, 4]. Practical video-based identification nevertheless requires localisation, feature extraction, identity matching, tracking, and temporal identity decisions to operate reliably together.

This thesis develops an end-to-end Cow Re-ID pipeline for two dairy farm scenarios, referred to as Farm A and Farm B. Here, end-to-end refers to a complete automated pipeline in which raw video footage is processed through a sequence of separately trained and

configured components to produce individual cow identity predictions. The datasets from both farms contain side-view cow images and videos, but the recording conditions are different. The Farm B recordings contain stronger occlusion from vertical metal bars, which motivated the use of a more specialised segmentation procedure.

The proposed system combines YOLO-based cow localisation and segmentation, DINOv3 embedding extraction, gallery-based cosine-similarity matching, BoT-SORT tracking, and track-level identity stabilisation. Reference images of each enrolled cow are converted into embeddings and combined into a cow-level gallery prototype. During video inference, query embeddings from segmented cow observations are compared with the stored cow-level prototypes using cosine similarity. If the similarity score and the matching evidence are strong enough, the corresponding cow ID is assigned. If the evidence is not sufficient, the cow remains unknown.

1.1 Aim and Objectives

The main aim of this thesis is to develop and evaluate an end-to-end computer-vision pipeline for individual cow re-identification from dairy farm videos. The main objectives are:

- to prepare and analyse farm-specific cow image and video datasets;
- to compare ResNet-50, MegaDescriptor, and DINOv3 for image-level Cow Re-ID;
- to develop suitable segmentation strategies for the two farm environments;
- to develop a gallery-based Cow Re-ID pipeline for video inference using DINOv3 embeddings, BoT-SORT tracking, confidence-based unknown handling, and track-level identity evidence; and
- to evaluate the proposed system using image-level and video-level experiments.

These objectives guide this thesis. Based on them, the following research questions are addressed:

RQ1: How do ResNet-50, MegaDescriptor, and DINOv3 compare for image-level Cow Re-ID in the two farm-specific datasets?

RQ2: How effectively can the proposed gallery-based video Cow Re-ID pipeline identify individual cows under the different recording conditions of Farm A and Farm B?

RQ3: How does video-based Cow Re-ID performance differ between the evaluated segmentation methods under the strong occlusion conditions of Farm B?

The main contribution of this thesis is the development and evaluation of an operational gallery-based video Cow Re-ID pipeline. Enrolled cow identities are represented by cow-level prototype embeddings, uncertain observations can remain unknown, and identity evidence is accumulated over tracked video frames before a cow ID is displayed. The pipeline integrates farm-specific segmentation, DINOv3-based gallery matching, BoT-SORT tracking, and track-level identity stabilisation, and is evaluated under two different farm conditions.

1.2 Thesis Structure

The rest of this thesis is organised into six chapters. Chapter 2 reviews the literature related to precision livestock farming, cow identification, vision-based Cow Re-ID, embedding-based methods, segmentation, tracking, and evaluation methods, and positions the proposed approach in relation to existing cattle Re-ID research. Chapter 3 describes the Farm A and Farm B datasets and the preprocessing procedures. Chapter 4 presents the proposed farm-specific Cow Re-ID methodology, including DINOv3 embedding extraction, segmentation, gallery construction, tracking, and identity matching. Chapter 5 describes the implementation and experimental setup. Chapter 6 presents and discusses the experimental results for image-level and video-level Cow Re-ID, segmentation, model comparison, and cross-farm transfer. Finally, Chapter 7 concludes the thesis and presents directions for future work.

2 Literature Review and Related Work

Individual cow identification is an important part of modern dairy farm monitoring. Many computer vision systems can detect that a cow is present in an image or video frame. However, for long-term farm management, detection alone is not enough. The system also needs to know which specific cow is being observed. This is important for monitoring health, behaviour, lameness, dirtiness, feeding activity, and welfare over time.

A targeted literature search was conducted to identify research directly related to vision-based cow re-identification, animal identification, embedding-based retrieval, segmentation, and video tracking. The literature and dataset search was performed using Google Scholar, IEEE Xplore, ScienceDirect, SpringerLink, arXiv, and the University of Bristol Research Data Repository. The main search terms included "cow re-identification," "cattle identification deep learning," "Holstein-Friesian cattle identification," "animal re-identification," "open-set cattle identification," "deep metric learning cattle," "cow identification dataset," "YOLO segmentation," "BoT-SORT tracking," and "Re-ID evaluation metrics."

The selection of literature focused on works that are directly related to the thesis pipeline. Papers were included when they discussed visual cow or cattle identification, animal Re-ID datasets, deep metric learning, query–gallery matching, foundation models, segmentation-based object extraction, multi-object tracking, or Re-ID evaluation metrics. Works that focused only on unrelated livestock sensing, non-visual identification without connection to Re-ID, or general deep learning without relevance to animal identification were not considered in detail. Priority was given to recent and highly relevant studies, while frequently cited foundational works were also included where appropriate. Particular attention was given to studies using cow/cattle datasets or practical farm video data.

Traditional cow identification methods such as ear tags, collars, and RFID are commonly used in dairy farms. These methods are useful, but they also have limitations. Ear tags can be damaged, dirty, or hidden from the camera view. RFID requires physical readers and usually works only when the cow is close to a specific location. Wearable devices also require maintenance and may increase cost in larger herds. For these reasons, vision-based cow re-identification has become an important research direction.

Cow re-identification, or Cow Re-ID, means recognising the same individual cow across different images, video frames, camera views, or days. It is different from cow detection. Detection answers the question: "Where is the cow?" Re-identification answers the question: "Which cow is this?" A practical system needs both. First, the cow must be detected or segmented. Then, useful visual features must be extracted and compared with known cow identities.

This thesis focuses on an end-to-end vision-based Cow Re-ID pipeline. The final system uses YOLO segmentation to isolate the cow body and reduce background noise. DINOv3 is used as the embedding extractor. Reference cow images are segmented and converted into cow-level prototype embeddings. During inference, query embeddings extracted from segmented cow crops in video frames are compared with these prototypes using cosine similarity. BoT-SORT is used to maintain tracking consistency across video frames. Finally, the system applies identity locking to stabilise the displayed cow ID and logging to store the identification results.

This chapter reviews the most relevant research areas for this thesis. It first discusses precision livestock farming and traditional cow identification methods in Sections 2.1 and 2.2. Vision-based Cow Re-ID and related datasets are reviewed in Sections 2.3 and 2.4. Deep learning, embedding-based Re-ID, and foundation models are discussed in Sections 2.5 and 2.6. The chapter then covers query–gallery matching, cow segmentation, video tracking, and evaluation methods in Sections 2.7–2.10. Finally, Section 2.11 presents the research gap and explains the position of this thesis.

2.1 Precision Livestock Farming and the Need for Cow Identity

Precision Livestock Farming uses sensors, cameras, and data-driven methods to monitor animals in a more automatic and continuous way. The goal is to support farmers with better information about animal health, welfare, productivity, and behaviour. Schillings et al. explain that precision livestock farming (PLF) technologies can help monitor animal health and welfare parameters continuously and automatically [1].

In dairy farms, camera-based monitoring is useful because it can observe animals without physical contact. Cameras can be installed in barns, walking passages, feeding areas, or milking areas. They can record cow movement, body appearance, behaviour, and interaction with the environment. However, camera-based monitoring becomes more useful when the recorded observations can be linked to the correct individual cow.

This is important because many animal monitoring tasks require individual-level information. For example, a lameness detection system may detect abnormal walking, but the result is only useful for farm management if the affected cow can be identified. Similarly, a dirtiness detection system may detect a dirty body region, but the farmer still needs to know which cow requires attention.

Schillings et al. [1] provide the broader Precision Livestock Farming background by showing that automated technologies can support continuous animal health and welfare monitoring. In this thesis, this background is connected to Cow Re-ID because camera-based monitoring becomes more useful when observations can be linked to the correct individual cow. Therefore, Cow Re-ID is considered an identity layer that can support other dairy farm AI systems, such as lameness detection, dirtiness detection, and long-term welfare monitoring.

2.2 Traditional Cow Identification Methods

Traditional cow identification methods include ear tags, RFID tags, collars, branding, and other forms of physical or manual identification. Among these, ear-tag and RFID-based identification are commonly discussed in cattle identification research [2]. These methods are widely used because they are known, practical, and already integrated into many farm management systems. RFID is especially useful in controlled areas such as milking stations and feeding stations, where the cow passes close to a reader.

However, these methods have limitations when the goal is visual monitoring from ordinary cameras. Ear tags may not be visible from all camera angles. They can also be hidden by the cow's head, covered by dirt, or blurred in motion. RFID tags require readers and infrastructure, so they do not directly identify cows in every part of the barn. Wearable devices can also be lost or damaged and may require battery management or maintenance.

Vision-based Cow Re-ID tries to reduce these limitations by using natural visual features of the cow. Hossain et al. reviewed cattle identification methods and showed that computer vision, machine learning, and deep learning are increasingly used for cattle detection and identification tasks [2]. For Holstein-Friesian cows, the black-and-white coat pattern is especially useful. The coat pattern covers a large body area and can be captured by side-view, top-view, or surveillance cameras. These natural markings can work as biometric features for cow identification [3, 4].

This does not mean that vision-based Re-ID must replace RFID or ear tags. In many real farms, these methods can work together. RFID can provide reliable identity at fixed stations, while vision-based Re-ID can provide identity information in camera-covered areas. In this

thesis, the focus is on vision-based identification because the goal is to identify cows from images and videos.

2.3 Vision-Based Cow Re-Identification

Vision-based Cow Re-ID uses image or video data to recognise individual cows. The model learns visual features that are related to identity. These features may include coat pattern, body markings, colour distribution, side-view appearance, and body shape. For Holstein-Friesian cows, coat pattern is one of the strongest identity cues because each cow has a distinctive black-and-white body pattern [3, 4].

A key work in this area is by Andrew et al. [3]. They studied visual identification of individual Holstein-Friesian cattle using deep metric learning. Their work used the natural coat patterns of Holstein-Friesian cattle for biometric identification. They developed a hands-free approach for detecting, localising, and identifying cattle from overhead images. This work is important because it showed that cows can be identified using visual appearance without relying only on tags or wearable devices.

One important part of Andrew et al.'s work is the open-herd setting [3]. In a closed-set classification setting, a model can only predict the cow IDs that were included during training. In contrast, an open-herd system should support new cows being added later without retraining the full model. This is very relevant for real farms because herd composition can change over time. New cows may be added, and some cows may leave the herd.

This idea is closely related to the approach used in this thesis. The trained Re-ID model is not used only as a fixed classifier. Instead, it is used as an embedding extractor. Reference images of cows are converted into gallery embeddings. A new cow crop from a video frame is also converted into an embedding. The system then compares the query embedding with the gallery embeddings. If the similarity is high enough, the cow ID is assigned. If the similarity is too low, the cow is treated as unknown.

Cow Re-ID is challenging because the same cow may look different in different situations. Lighting conditions can change during the day. Camera angle and cow pose can vary. Cows can be partially occluded by other cows, gates, or farm structures. Dirt and movement can change the visible appearance. The cow may also be only partly visible in the frame. Therefore, a practical Cow Re-ID system needs more than a good classifier. It needs reliable cow localisation, clean cow crops, robust embeddings, gallery matching, temporal tracking, and identity consistency checking.

2.4 Cow Re-ID Datasets and Related Research

Public datasets are important for Cow Re-ID research because they allow models and methods to be trained and compared. The development of Cow Re-ID datasets also shows how the research field has moved from image-based identification towards video-based and multi-camera identification.

Andrew et al. introduced the OpenCows2020 dataset together with their work on visual identification of Holstein-Friesian cattle [3, 5]. OpenCows2020 is important because it contains two relevant components: detection/localisation and identification [5]. Detection/localisation refers to finding where the cow is located in the image, while identification refers to assigning the correct individual cow ID. This matches the structure of a practical Cow Re-ID system. First, the cow must be detected or localised. Then, the cow must be identified. This separation is also important in the present thesis, where YOLO segmentation is used before Re-ID matching.

Gao et al. introduced the Cows2021 dataset for video identification of individual Holstein-Friesian cattle [4]. The dataset contains 10,402 RGB images with localisation and identity labels, as well as 301 videos from the same herd. The images were captured from top-down in-barn views and show the distinctive black-and-white coat patterns of the cows. The authors also explored self-supervised cow identity learning by using video tracklets, where frames from the same tracklet were treated as visually related samples of the same cow. This is important because manual identity labelling is one of the major challenges in animal Re-ID.

In a later work, Gao et al. proposed "Label a Herd in Minutes" [6]. This work focused on reducing manual labelling effort for cattle identification. The authors combined self-supervision, metric learning, clustering, and active learning. Their approach used automatically generated tracklets from farm CCTV footage and required only limited human labelling effort. This is relevant because a practical cow identification system should not require full manual labelling of every frame.

More recently, Yu et al. introduced MultiCamCows2024 for multi-view Cow Re-ID on a working farm [7]. The dataset contains 101,329 images of 90 cows captured by three ceiling-mounted cameras over seven days. It also includes the original CCTV footage and provides supervised and self-supervised baselines. This dataset is important because it represents a more realistic farm setting. In real farms, cows may move through areas covered by different cameras, and the system must handle viewpoint changes, different backgrounds, and different lighting conditions.

In a recent study, Yu et al. proposed an automated Re-ID approach for Holstein-Friesian cattle in dense groups [8]. Their method uses a detect–segment–identify pipeline in which OWLv2 is first used to localise individual cows and SAM2 is then used to obtain instance segmentations. The segmented cow images are subsequently used for re-identification with an unsupervised contrastive learning approach. The method was evaluated using CCTV data collected over nine days on a working dairy farm. The authors reported an average Re-ID accuracy of 94.82% across the nine-day evaluation. This work is particularly relevant because it demonstrates an automated segmentation-based Re-ID approach for individual cattle when multiple cows are closely grouped or occluded.

Most of these datasets focus on Holstein-Friesian cattle because their black-and-white coat patterns are highly distinctive. However, vision-based Re-ID is not limited only to Holstein-Friesian cattle. The same general approach can also work for other cow categories when they have visible identity-related markings, colour patterns, or body appearance differences. In this thesis, the system is designed for side-view farm images, where the cow body and coat pattern are visible. This makes the approach suitable for cows with visible coat-pattern differences.

Other cattle identification studies have also explored different biometric features. Hossain et al. reviewed machine learning and deep learning methods for cattle identification and reported that muzzle prints and coat patterns are among the common distinguishing features used in cattle identification research [2]. Zhao and Lian proposed compact loss for visual identification of cattle in the wild, combining classification and metric learning ideas to improve feature separability [9].

These works are directly related to this thesis. They show that cow identity can be learned from visual features, especially coat patterns. They also show that detection, segmentation, video-based identification, self-supervised learning, and open-set identification are important research directions. Recent work has additionally demonstrated the integration of multiple components within automated cattle Re-ID pipelines. However, practical video-based Cow Re-ID still faces challenges under farm-specific conditions, including occlusion, background and lighting variation, cows that are not enrolled in the reference gallery, and unstable identity predictions across video frames. These practical challenges and the position of this thesis are discussed further in Section 2.11.1.

2.5 Deep Learning and Embedding-Based Cow Re-ID

Deep learning is widely used for visual recognition because it can learn identity-relevant features directly from image data. In Cow Re-ID, these features may represent coat patterns,

body markings, colour distributions, and other visual characteristics that distinguish one cow from another. Depending on how the model is trained and used, cow identification can be formulated either as a closed-set classification problem or as an embedding-based retrieval problem.

In closed-set classification, the model predicts one cow identity from a fixed set of classes available during training. ResNet-50, introduced by He et al. [10], is a widely used convolutional neural network based on residual learning. Its residual connections make it easier to train deeper networks by allowing information to pass through shortcut connections [10]. In this thesis, ResNet-50 was used as a conventional CNN-based baseline and trained using a fixed set of cow identities. It therefore provided a useful reference for evaluating the performance of the other models under closed-set identification conditions.

However, the classification output of ResNet-50 is limited to the cow identities represented during training. When a new cow is introduced, a conventional closed-set classifier cannot directly assign an identity that was not represented among its training classes. Its classification layer would normally need to be modified, and the model would need to be retrained or further adapted. In this thesis, the adapted ResNet-50 model also produces a 256-dimensional embedding representation, and these learned features were evaluated using image-level retrieval metrics, as presented in Section 6.2. For the experimental comparison, ResNet-50 was used as a conventional CNN baseline for both image-level classification and retrieval evaluation.

Embedding-based Re-ID provides a more flexible approach. Instead of directly predicting a fixed cow class, the model converts an image into a feature vector called an embedding. A suitable embedding model should place images of the same cow close together in the embedding space while separating images of different cows. A query embedding can then be compared with stored gallery embeddings to identify the most visually similar cow.

Deep metric learning is commonly used to create such discriminative embedding spaces. Andrew et al. used deep metric learning for Holstein-Friesian cattle identification and proposed a SoftMax-based reciprocal triplet loss [3]. Their work demonstrated that embedding-based identification can support an open-herd setting in which new cattle can be enrolled without retraining the complete identification model. Zhao and Lian proposed compact loss for visual cattle identification under an open-set protocol [9]. Their method combines SoftMax-nB with metric-learning objectives, including triplet loss and tight loss, to increase the separation between different cattle identities and reduce variations among images of the same identity.

The final Cow Re-ID system developed in this thesis follows this embedding-based approach. ResNet-50, MegaDescriptor, and DINOv3 were compared using both image-level

classification and retrieval evaluation. ResNet-50 served as the conventional CNN baseline, while MegaDescriptor and DINOv3 were included as alternative pretrained representation models developed for animal Re-ID and large-scale visual representation learning, respectively. The image-level experimental results are presented in Section 6.2, while the selection of the embedding model used in the final video-based Cow Re-ID pipeline is discussed in Section 6.2.3. MegaDescriptor and DINOv3 are discussed in the following section.

2.6 Foundation Models for Animal Re-ID

Recent computer vision research has moved towards foundation models. These models are trained on large and diverse datasets and can provide strong visual features for many tasks. Instead of training a model from scratch for each small dataset, a foundation model can be adapted or used as a feature extractor.

In animal Re-ID, WildlifeDatasets and MegaDescriptor are important works. Čermák et al. introduced WildlifeDatasets as an open-source toolkit for animal re-identification [11]. The toolkit provides access to animal Re-ID datasets, preprocessing tools, evaluation support, and model fine-tuning. The same work introduced MegaDescriptor, a foundation model for individual animal Re-ID across different species [11]. MegaDescriptor is relevant to this thesis because it was one of the tested models in the project.

DINO-style models are also relevant because they learn strong visual features using self-supervised learning. Caron et al. introduced DINO and studied self-supervised Vision Transformers [12]. Their work showed that self-supervised Vision Transformer features can contain useful visual information. DINOv2 extended this direction by learning robust visual features without supervision and by scaling training using large curated data [13]. DINOv3 further developed this family as a self-supervised vision foundation model designed to produce high-quality visual representations across different visual tasks [14].

In this thesis, DINOv3 is used as the main embedding extractor. It was selected because it showed the strongest overall performance among the tested models across the image-level classification and retrieval evaluation and produced suitable embeddings for query–gallery-based Cow Re-ID.

It is important to state this carefully. This thesis does not claim that DINOv3 is the best model for all animal Re-ID problems. The claim is more specific: in this project, DINOv3 showed the strongest overall performance among the tested models and was suitable for the final gallery-based Re-ID pipeline. The following subsections give a more focused description of MegaDescriptor and DINOv3 because both models were tested in this thesis and are directly related to embedding-based Cow Re-ID.

2.6.1 MegaDescriptor

MegaDescriptor is a foundation feature descriptor designed for animal re-identification. The main goal is to convert an input animal image into a fixed-length embedding vector so that images of the same individual have embeddings that are close to each other, while images of different individuals are far apart in the embedding space. This makes MegaDescriptor useful for Re-ID tasks such as image retrieval and identity matching [11].

MegaDescriptor was introduced as part of the WildlifeDatasets toolkit [11]. The WildlifeDatasets work describes MegaDescriptor as a strong animal Re-ID model and evaluates it across multiple animal Re-ID datasets [11]. It is relevant to this thesis because cow identification is also an individual animal Re-ID problem.

As described by Čermák et al., MegaDescriptor is based on Swin Transformer models optimised with ArcFace loss [11]. The Swin Transformer is a hierarchical vision transformer architecture that uses shifted-window attention. This design keeps computation efficient while still learning strong visual representations [15]. The resulting model produces a deep embedding vector that can be used for identity matching [11].

MegaDescriptor is trained using a metric-learning style objective. This means that the training goal is to learn a feature space where identities are well separated. The WildlifeDatasets work uses ArcFace loss for MegaDescriptor training [11]. ArcFace is an additive angular margin loss that encourages stronger separation between identities in the embedding space [16].

During training, ArcFace loss is used through a classification layer to optimise the embedding space [16]. This layer is required for computing the training loss, but it is not used for nearest-neighbour identity matching during inference. During inference, the model's output embedding is used for matching. Reference images are first passed through MegaDescriptor to create precomputed gallery embeddings. A query image is then converted into a query embedding, which is compared with the precomputed gallery embeddings using cosine similarity. In the WildlifeDatasets evaluation protocol, MegaDescriptor uses a 1-nearest-neighbour classifier in the deep embedding space for identity matching [11].

In this thesis, the pretrained MegaDescriptor backbone was adapted using the same 256-dimensional embedding layer, batch-normalisation layer, and linear cow-ID classifier used for the other evaluated backbones. The complete adapted network was fine-tuned using cross-entropy loss. Therefore, the ArcFace objective described above refers to the original MegaDescriptor training procedure and was not reimplemented in this thesis.

In this thesis, MegaDescriptor was evaluated because it was developed specifically for individual animal Re-ID. Its experimental performance is compared with ResNet-50 and DINOv3

in the evaluation chapter. Figure 2.1 shows the high-level architecture of MegaDescriptor, including the Swin Transformer backbone, feature extraction, and embedding generation used for identity matching.

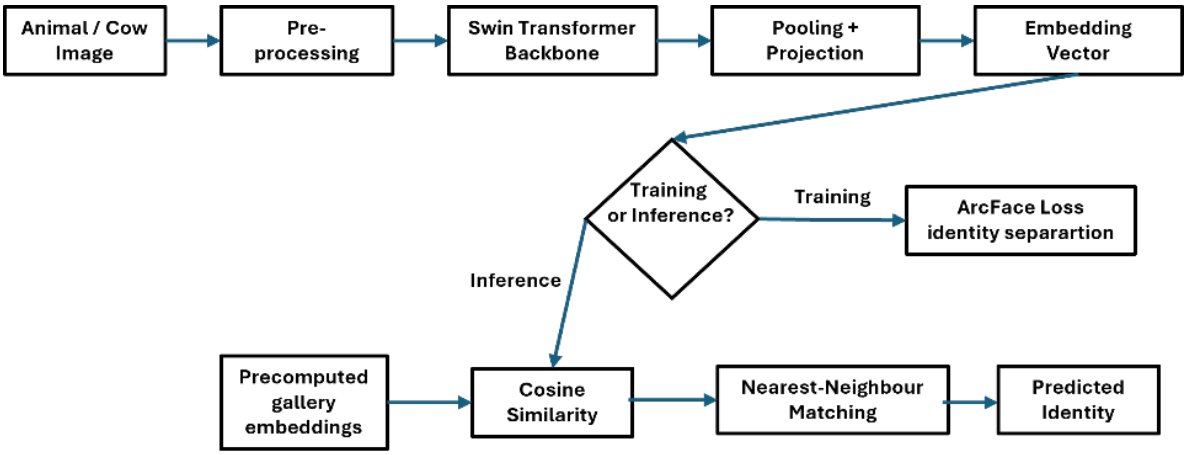


Figure 2.1: High-level architecture of MegaDescriptor, based on Čermák et al. [11].

2.6.2 DINOv3

DINOv3 belongs to the DINO family of self-supervised vision foundation models. The original DINO method was introduced by Caron et al. and uses self-distillation without labels to learn strong visual representations [12]. DINOv2 and DINOv3 are later models in this family that further improve general-purpose visual feature learning [13, 14].

DINO-style methods are commonly used with Vision Transformer backbones. Vision Transformer, or ViT, processes an image by dividing it into fixed-size patches. These patches are flattened and projected into token embeddings. The tokens are then processed by transformer encoder blocks that use self-attention and feed-forward layers [17].

In DINO-style self-supervised learning, a student network learns by matching the output of a teacher network. The teacher network is updated more slowly, for example using an exponential moving average of the student weights. This teacher–student self-distillation framework helps the model learn useful visual representations without manual class labels during pretraining [12]. Therefore, the teacher–student idea should be understood as part of the DINO family, not as something first introduced only by DINOv3.

DINOv3 was introduced by Siméoni et al. in 2025 [14]. One important improvement in DINOv3 is Gram anchoring. This regularisation technique helps maintain stable patch-level features during long training schedules [14]. DINOv3 also applies post-hoc strategies, such as high-resolution adaptation and model distillation. These strategies improve the flexibility of the model with respect to image resolution and model size [14].

In this thesis, pretrained DINOv3 models were fine-tuned and evaluated separately using labelled cow-ID data from the two farm scenarios. During inference, the corresponding farm-specific fine-tuned model is used to extract embedding vectors. The extracted cow embedding is compared with stored cow-level prototype embeddings using cosine similarity. This makes DINOv3 suitable for gallery-based Cow Re-ID, where the goal is to match a detected cow against known cow identities. Figure 2.2 shows the high-level DINO-style teacher–student self-distillation framework and the main improvements introduced in DINOv3.

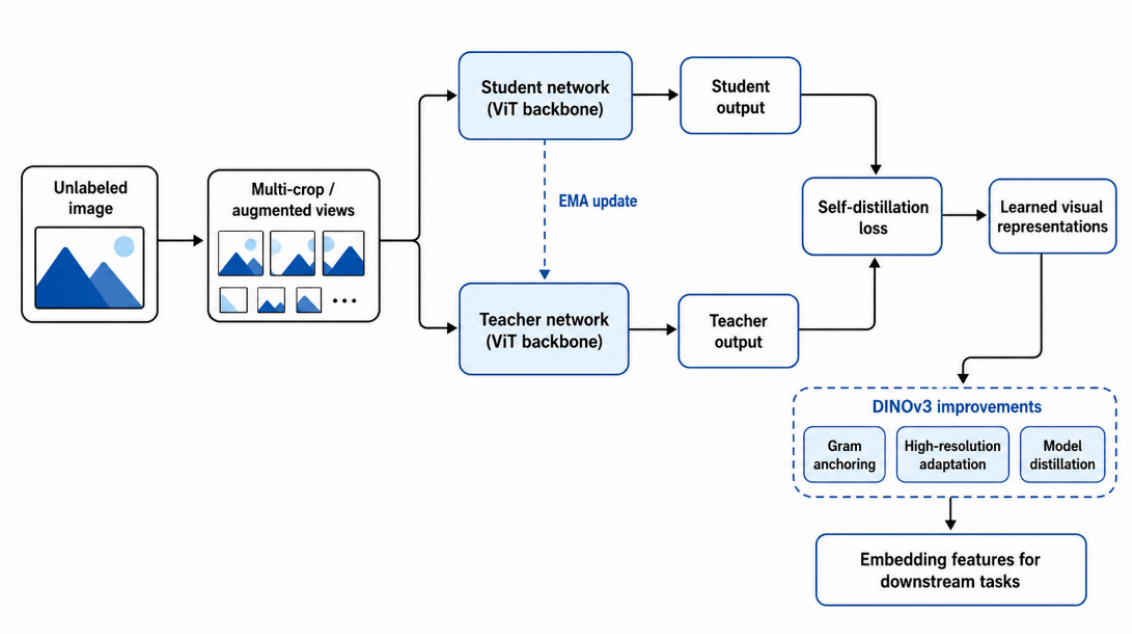


Figure 2.2: High-level DINO-style teacher–student self-distillation and DINOv3 improvements, based on DINO [12] and DINOv3 [14].

After introducing the embedding models relevant to this thesis, the next section explains how these embeddings are used for identity assignment through query–gallery matching and open-set identification.

2.7 Query–Gallery Matching and Open-Set Identification

Query–gallery matching is a common structure in Re-ID systems [18, 19]. The gallery contains reference images or embeddings of known identities, while the query is a new image that must be identified. The system compares the query embedding with the gallery embeddings and selects the closest identity. This retrieval-based structure is widely used in person Re-ID and provides the basis for standard Re-ID metrics such as Rank-k accuracy and mean average precision [18, 19]. The same query–gallery structure can be applied to Cow Re-ID, where the identities are individual cows instead of people.

Query–gallery matching can be used in both closed-set and open-set settings. In an open-set setting, the system must additionally be able to reject a query when no sufficiently confident gallery identity is available [20].

This approach is useful for cow identification because a farm may need to add new cows over time. In the proposed system, reference images of new cows can be collected and processed. YOLO segmentation is used to isolate the cow body, and the resulting cow-only crops are passed through the corresponding farm-specific fine-tuned DINOv3 model to extract image-level embeddings. For each cow ID, these embeddings are averaged and L2-normalised to create a cow-level prototype embedding, which is stored in the inference gallery. This process allows the gallery to be updated without retraining the Re-ID model. It is also related to the open-herd idea discussed by Andrew et al., where new cattle can be identified without retraining the complete system [3].

During inference, the corresponding farm-specific preprocessing procedure and fine-tuned DINOv3 model are used to create a query embedding from each valid cow crop. The query embedding is compared with the cow-level prototype embeddings stored in the inference gallery using cosine similarity. A frame-level match is accepted only when the best similarity score and the best-to-second-best similarity margin satisfy their configured thresholds. The accepted match then contributes identity evidence to the current track. A cow ID is displayed only after the accumulated track-level evidence satisfies the identity-locking requirements. Otherwise, the track remains unknown.

The unknown decision is important in real farm use because not every detected cow is necessarily enrolled in the gallery. A closed-set classifier normally selects one identity from the known classes, even when the cow is not actually part of those classes. This can lead to wrong identification. In contrast, open-set recognition aims to reject samples that do not belong to the known enrolled identities [20]. Therefore, gallery-based matching can use a similarity threshold. If the best match is not confident enough, the system can mark the cow as unknown instead of forcing an incorrect ID.

The similarity threshold must be chosen carefully. A low threshold may identify more cows, but it can also increase false matches. A high threshold may reduce false matches, but more cows may remain unknown. In this thesis, the margin between the best and second-best similarity scores is also used as an additional confidence cue. If two cow IDs have very similar scores, the system should be careful before assigning an identity. This is intended to reduce ambiguous decisions when two cows have similar coat patterns or when the query crop is unclear.

For video-based inference, it is also risky to assign an ID from one frame only. A single frame may be blurred, occluded, or poorly segmented. Therefore, this thesis uses track-level

identity evidence together with BoT-SORT tracking [21]. The system waits for repeated confident matches before locking and displaying a cow ID. This is intended to make the final identity decision more stable than relying on a single frame.

2.8 Cow Detection and YOLO Segmentation

Before a cow can be identified, the system must first locate the cow in the image or video frame. Object detection is the task of finding objects and usually gives bounding boxes. YOLO is one of the most well-known real-time object detection approaches. Redmon et al. introduced YOLO as a unified object detection method that predicts bounding boxes and class probabilities directly from the full image in one evaluation [22].

Object detection is useful, but bounding boxes are not always enough for Cow Re-ID. A bounding-box crop may contain a lot of background, such as the barn floor, wall, gates, shadows, feeding structures, or other cows. These background regions are not part of the cow identity. If they remain in the image, they may add noise to the embedding and reduce matching quality.

Instance segmentation gives a more precise object region than a bounding box. Mask R-CNN is a well-known instance segmentation method that detects objects and also predicts a segmentation mask for each instance [23]. Modern YOLO models also support instance segmentation. The Ultralytics YOLO documentation describes instance segmentation as identifying individual objects in an image and segmenting them from the rest of the image [24]. YOLO11 also supports multiple computer vision tasks, including object detection, instance segmentation, classification, pose estimation, and oriented object detection [25].

This thesis uses YOLO segmentation to isolate the cow body from the background. This is an important design choice because Cow Re-ID depends strongly on the visible coat pattern. By applying segmentation, the system can focus more on the cow body and less on the farm environment.

For each farm scenario, the same farm-specific segmentation model was used in both gallery construction and video inference to retain the visible cow body and suppress background information. This consistency was important because the gallery and query embeddings should be created from visually comparable cow crops. If gallery embeddings are created from full images containing background, while video embeddings are created from segmented cow crops, the resulting gallery and query distributions may differ. Similarly, if gallery images are segmented but video observations are not, the query embeddings may contain background information that is not represented in the gallery. Therefore, applying the same farm-specific

segmentation approach in both stages helps reduce this source of inconsistency and makes the gallery and query embeddings more comparable.

Segmentation can still fail in difficult cases. A mask may cover only part of the cow, particularly under strong occlusion. In addition, visual similarity between the coat patterns of different cows can lead to misidentification in appearance-based Cow Re-ID [7]. Distinctive coat patterns provide important visual cues for identifying individual Holstein-Friesian cattle [3, 4]. However, when the mask is accurate, segmentation can provide a cleaner Re-ID input by reducing background information compared with raw bounding-box cropping.

2.9 Video-Based Re-ID and BoT-SORT Tracking

Image-based Re-ID identifies a cow from a single image. Video-based Re-ID is particularly relevant for real farm monitoring because cows can be observed continuously over multiple video frames. However, video creates additional challenges. A cow can move, turn, overlap with another cow, leave the frame, re-enter the frame, or become partially visible. The Re-ID score can also fluctuate from frame to frame.

Multi-object tracking helps address this problem. A tracker follows detected objects across frames and assigns a temporary track ID. The tracker does not necessarily know the biological identity of the cow. Its main purpose is to determine whether the same detected object continues across frames. The Re-ID module then assigns a predicted biological cow identity to the track.

BoT-SORT is a multi-object tracking method proposed by Aharon et al. [21]. It combines motion information, appearance information, camera-motion compensation, and an improved Kalman filter state representation. BoT-SORT was originally developed for pedestrian tracking, but the tracking idea is useful in cow videos as well because cows also need to be followed across frames.

In this thesis, BoT-SORT is used as the tracking component. It maintains track continuity for detected cows in the video. DINOv3-based gallery matching is used as the cow identity component. This separation is important. BoT-SORT helps keep the same moving cow connected across frames, but it does not by itself know whether the cow is Cow 425, Cow 433, or another cow. The Re-ID module assigns the biological cow ID by comparing embeddings with the gallery.

Tracking is important because frame-level Re-ID can be unstable. If the model predicts an ID independently for every frame, the displayed ID may switch due to one weak crop or one bad frame. A cow may also be partly hidden or blurred. By using track-level evidence, the system can wait for repeated confident matches before displaying the identity. This is

intended to reduce unstable frame-level identity changes and make the displayed output more reliable.

The final system also uses delayed identity locking. Instead of immediately showing an ID after one high similarity score, the system accumulates evidence over multiple frames. Once the accumulated evidence satisfies the configured identity-locking requirements, the cow ID is displayed. If the evidence is insufficient or becomes inconsistent, the system can keep the track unknown or reset the identity. This conservative decision strategy allows uncertain tracks to remain unknown rather than forcing an identity prediction.

2.10 Evaluation Metrics for Cow Re-ID

Cow Re-ID can be evaluated in different ways. For image-level model evaluation, closed-set classification accuracy is useful. It measures how often the model predicts the correct cow ID from a fixed set of known classes. For N test images, classification accuracy is calculated as

$$\text{Accuracy} = \frac{N_{\text{correct}}}{N}, \tag{2.1}$$

where N_{correct} is the number of correctly classified test images. This is useful for comparing the classification performance of the tested models.

Since cow-ID classification is a multi-class problem, macro precision, macro recall, and macro F1-score are used in addition to classification accuracy to summarise performance across all cow identities. Precision and Recall are first calculated separately for each cow identity c as

$$P_c = \frac{TP_c}{TP_c + FP_c}, \qquad R_c = \frac{TP_c}{TP_c + FN_c}, \tag{2.2}$$

where TP_c, FP_c, and FN_c denote the true-positive, false-positive, and false-negative predictions for cow identity c. The F1-score for each identity is

$$F1_c = 2\frac{P_c R_c}{P_c + R_c}. \tag{2.3}$$

The macro measures are calculated by averaging the corresponding values across all cow identities, where C denotes the total number of cow identities:

$$\text{Macro Precision} = \frac{1}{C}\sum_{c=1}^{C} P_c, \qquad \text{Macro Recall} = \frac{1}{C}\sum_{c=1}^{C} R_c, \tag{2.4}$$

$$\text{Macro F1} = \frac{1}{C}\sum_{c=1}^{C} F1_c. \tag{2.5}$$

This gives equal importance to each cow identity, regardless of how many test images are available for that cow.

However, closed-set classification metrics alone do not fully describe gallery-based Re-ID performance. Retrieval metrics are therefore also important. Rank-k accuracy measures how often the correct cow identity occurs within the first k retrieved gallery matches. For N_q query images, it is calculated as

$$\text{Rank-}k = \frac{1}{N_q}\sum_{q=1}^{N_q} \mathbf{1}(r_q \leq k), \tag{2.6}$$

where r_q is the rank position of the correct identity for query q, and $\mathbf{1}(\cdot)$ is an indicator function which has a value of 1 when the correct identity appears within the first k retrieved matches; otherwise, it is 0. In this thesis, Rank-1 and Rank-5 are reported.

Mean average precision (mAP) evaluates the overall quality of the retrieval ranking. It is calculated by averaging the average precision (AP) over all queries:

$$\text{mAP} = \frac{1}{N_q}\sum_{q=1}^{N_q} \text{AP}_q, \tag{2.7}$$

where AP_q is the average precision obtained for query q.

These retrieval metrics are commonly used in Re-ID research because they evaluate the ranking of gallery candidates rather than only one fixed classification output [18, 19]. In this thesis, Rank-1, Rank-5, and mAP are used to compare the 256-dimensional embedding representations produced by ResNet-50, MegaDescriptor, and DINOv3.

Video inference requires additional evaluation because the final practical result is based on cow identities recovered from video tracks rather than on single test images. For Farm A, identification performance is expressed as an identification rate:

$$\text{Identification Rate} = \frac{N_{\text{correct}}}{N_{\text{expected}}} \times 100\%, \tag{2.8}$$

where N_{correct} is the number of correctly identified target cases and N_{expected} is the corresponding number of expected cases. The same proportion is used to describe recovery across the daytime and night-time Farm A experiments.

For Farm B, RFID timestamps provide an external temporal reference for the video-level evaluation. The resulting predictions are grouped into true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) according to the RFID matching procedure described in Section 5.7.6. From these values, Precision, Recall, F1-score, Specificity, and Accuracy are calculated as

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN}, \tag{2.9}$$

$$\text{F1} = 2\frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}, \tag{2.10}$$

$$\text{Specificity} = \frac{TN}{TN + FP}, \tag{2.11}$$

and

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}. \tag{2.12}$$

Precision describes the reliability of positive identifications, Recall describes how many expected positive cases are recovered, F1-score balances Precision and Recall, Specificity measures correct rejection of negative cases, and Accuracy gives the overall proportion of correct decisions.

Therefore, the evaluation in this thesis includes both image-level and video-level measures. Image-level evaluation uses classification accuracy, macro precision, macro recall, macro F1-score, Rank-1, Rank-5, and mAP. Video-level evaluation uses identification rates for Farm A and RFID-supported Precision, Recall, F1-score, Specificity, and Accuracy for Farm B.

2.11 Research Gap and Position of This Thesis

The previous sections reviewed the main research areas related to vision-based Cow Re-ID. Based on this literature, the following subsections identify the main research gap and explain how this thesis is positioned in relation to the existing work.

2.11.1 Research Gap

The reviewed literature shows that vision-based cattle identification is feasible in image-based, video-based, open-herd, and multi-camera settings. Andrew et al. demonstrated visual identification of Holstein-Friesian cattle using deep metric learning and evaluated identification in an open-herd setting [3]. Cows2021 and "Label a Herd in Minutes" further explored video-based and self-supervised cattle identification [4, 6], while MultiCamCows2024 extended this research towards multi-camera identification under working-farm conditions [7]. More recently, Yu et al. demonstrated an automated detect–segment–identify pipeline for cattle Re-ID in densely grouped working-farm video [8].

These studies show that the individual components of cattle Re-ID, and even their integration into automated video-based pipelines, have already received significant research attention. However, practical video-based identification also requires an operational mechanism for deciding when a detected and tracked cow can be assigned an enrolled identity and when the available evidence is not sufficiently reliable.

The present thesis addresses this practical challenge by focusing on reliable identity assignment in continuous farm video, including the rejection of uncertain matches and the accumulation of identity evidence over tracked video observations. The specific position of the proposed gallery-based pipeline in relation to existing work is described in the following subsection.

2.11.2 Position of This Thesis

Based on the reviewed literature, this thesis is positioned as an applied end-to-end, gallery-based Cow Re-ID system for dairy farm video monitoring. It builds on previous research in visual cattle identification, animal Re-ID, self-supervised visual representations, instance segmentation, gallery-based matching, and multi-object tracking.

The proposed system is designed around two connected stages: gallery construction and continuous video inference. During gallery construction, multiple reference images of each enrolled cow are converted into embeddings and combined into a cow-level prototype. During video inference, segmented cow observations are tracked across frames and their query embeddings are compared with the stored prototypes using cosine similarity.

The system does not force an enrolled identity from every observation. A cow can remain unknown when the matching evidence is insufficient. Furthermore, the biological cow identity is not displayed based on a single frame prediction. Instead, identity evidence is accumulated over multiple tracked frames and the cow ID is displayed only after the configured identity-locking requirements are satisfied.

To place the proposed approach in the context of existing cattle and animal Re-ID research, Table 2.1 summarises the main related approaches together with their research focus, methods, datasets, and evaluation settings.

Table 2.1: Summary of related approaches, models, datasets, and evaluation settings

Study	Main focus	Model / method	Dataset / data	Evaluation
Andrew et al. [3]	Holstein-Friesian cattle identification	CNN-based detection and deep metric learning with SoftMax-based reciprocal triplet loss	OpenCows2020	Identification accuracy and open-herd evaluation
Gao et al. [4]	Video-based cow identification	Self-supervised metric learning using video tracklets	Cows2021	Top-k identification accuracy and clustering-based evaluation
Gao et al. [6]	Reducing manual labelling effort	Self-supervision, metric learning, clustering, and active learning	Farm CCTV tracklets	Identification accuracy and clustering-quality evaluation
Yu et al. [7]	Multi-camera Cow Re-ID	Supervised and self-supervised identification baselines	MultiCamCows2024	Single-image identification accuracy and multi-camera evaluation
Yu et al. [8]	Automated Re-ID of densely grouped Holstein-Friesian cattle	OWLv2 localisation, SAM2 segmentation and tracking, and unsupervised contrastive learning	Nine days of working-farm CCTV	Detection/segmentation performance and kNN Re-ID accuracy on unseen-day data
Zhao and Lian [9]	Open-set cattle identification	Compact loss with SoftMax-nB, triplet loss, and tight loss	Cattle images under open-set conditions	Open-set identification performance
Čermák et al. [11]	General animal Re-ID	WildlifeDatasets toolkit and MegaDescriptor foundation model	Multiple animal Re-ID datasets	Reference–query identification accuracy across multiple animal Re-ID datasets
This thesis	Gallery-based video Cow Re-ID with unknown handling and temporal identity stabilisation	YOLO segmentation, DINOv3 embeddings, cow-level prototype matching, BoT-SORT tracking, and delayed identity locking	Farm-specific side-view cow images and videos	Image-level classification and retrieval metrics, Farm A identification rates, and RFID-supported Farm B video metrics

The comparison in Table 2.1 shows that recent cattle Re-ID research has progressed from image-based identification towards video-based, multi-camera, open-set, and automated identification settings. Recent work has also demonstrated integrated detection, segmentation, tracking, and embedding-based identification under working-farm conditions. The present thesis addresses a complementary operational setting by using cow-level prototype embeddings for gallery matching, allowing uncertain observations to remain unknown, and accumulating identity evidence over tracked video frames before displaying a cow ID.

Therefore, the main position of this thesis is an operational video-based Cow Re-ID pipeline that combines farm-specific segmentation, prototype-based gallery matching, explicit unknown decisions, and track-level identity stabilisation. The complete system is evaluated under two different farm conditions to examine its performance under different recording and occlusion conditions.

2.12 Summary

This chapter reviewed the main research areas related to vision-based Cow Re-ID. Precision livestock farming provides the broader application context, while traditional identification methods such as RFID and ear tags have limitations for continuous camera-based identification. Previous research shows that individual cows can be identified using visual features such as coat patterns and body markings.

The reviewed literature also shows the development of cattle Re-ID from image-based identification towards video-based, open-set, multi-camera, and automated identification under working-farm conditions. Deep learning, embedding-based retrieval, foundation models, segmentation, query–gallery matching, and multi-object tracking provide the main technical foundations for these systems.

Recent studies have demonstrated that multiple components can be integrated within automated cattle Re-ID pipelines. Based on this literature, the present thesis focuses on an operational gallery-based video Re-ID setting in which enrolled cows are represented by prototype embeddings, uncertain observations can remain unknown, and identity evidence is accumulated over tracked video frames before a cow ID is displayed. The next chapter describes the two farm-specific datasets and the preprocessing procedures used for image-level evaluation, inference-gallery construction, and video-based Cow Re-ID.

3 Dataset Description and Preprocessing

This chapter describes the datasets and preprocessing steps used in this thesis. Before discussing the datasets in detail, the overall architecture of the farm-specific Cow Re-ID pipeline is introduced. The system was developed and evaluated using two farm-specific scenarios: Farm A and Farm B. Both farms provided side-view cow image and video data, but the visual conditions and segmentation requirements were different. Therefore, each farm was treated as a separate experimental scenario.

The following sections first introduce the overall architecture of the farm-specific Cow Re-ID pipeline in Section 3.1. The dataset sources and the two farm-specific scenarios are described in Section 3.2, with the Farm A and Farm B datasets presented in Sections 3.2.1 and 3.2.2, respectively. Section 3.3 then describes the preprocessing procedures. Image extraction and dataset organisation are explained in Section 3.3.1, followed by the train, validation, and test split in Section 3.3.2 and the query–gallery split for image-level evaluation in Section 3.3.3. Final inference-gallery preparation and segmentation preprocessing are presented in Section 3.3.4, while the video data used for final inference are described in Section 3.3.5. Finally, Section 3.3.6 explains preprocessing during video inference and query-embedding extraction.

3.1 Overall Architecture of the Farm-Specific Cow Re-ID Pipeline

The overall architecture of the farm-specific Cow Re-ID pipeline is shown in Figure 3.1. The complete pipeline contains two main stages. The first stage includes image-level model training, shown in the blue blocks, and image-level evaluation, shown in the orange block. In this stage, cow images are organised by cow ID, split into training, validation, and test sets, and used to train and evaluate the farm-specific ResNet-50, MegaDescriptor, and DINOv3 models. The train, validation, and test split is performed inside each cow ID folder. Therefore, the same cow IDs appear in all three splits, but with different images. Classification performance is evaluated using the held-out test images, while query and gallery sets for image-level retrieval evaluation are created from the test split. ResNet-50, MegaDescriptor, and DINOv3

were trained and compared at image level, while DINOv3 was used as the embedding model in the final video-inference pipeline.

The second stage is the final video-inference pipeline, shown in the green blocks. In this stage, the gallery-enrolment procedure differed slightly between the two farms. For Farm A, the cow IDs included in the final inference gallery were not used during training, validation, or testing. For Farm B, the cow IDs included in the final inference gallery were also present in the training, validation, and test sets.

For both farms, the gallery images were separate from the videos used for inference. The inference videos also contained some cows that were not represented in the corresponding gallery. The gallery images were segmented, converted into DINOv3 embeddings, and averaged into cow-level prototype embeddings. During video inference, cows were detected, tracked, segmented, converted into query embeddings, and matched against the gallery prototypes using cosine similarity, shown in the purple block. Finally, the red block represents the final output, where the system displays either a matched cow ID or an unknown decision.

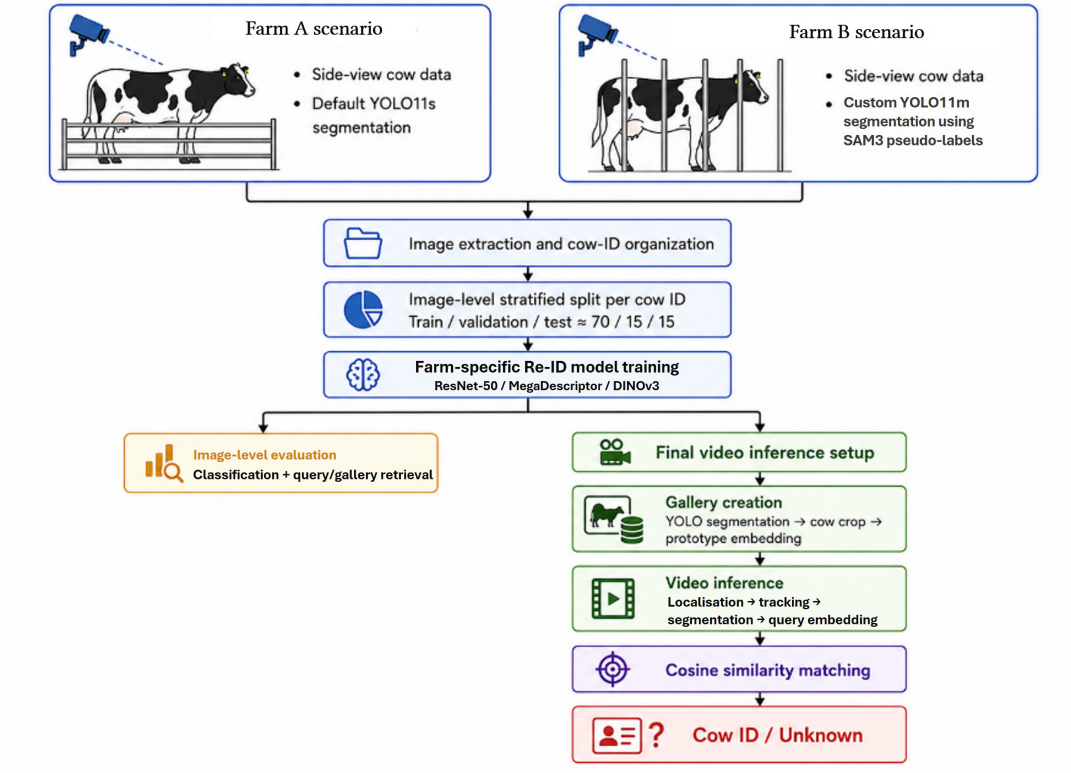


Figure 3.1: Overall architecture of the farm-specific Cow Re-ID pipeline.

The two farm scenarios use the same general Re-ID concept, but different segmentation strategies. For Farm A, the default YOLO11s segmentation model was used because the segmentation conditions were less challenging than in Farm B. For Farm B, a custom YOLO11m

segmentation model trained using SAM3-generated pseudo-labels was used because stronger vertical metal-bar occlusion made default segmentation less reliable.

3.2 Dataset Sources and Farm Scenarios

The datasets used in this thesis were provided by the Fraunhofer-Institut für Graphische Datenverarbeitung (Fraunhofer IGD) and were collected from two farms. For confidentiality, the farms are referred to as Farm A and Farm B. The datasets were used with permission from Fraunhofer IGD and the respective farm operators.

The datasets from both farms contained side-view camera data. This camera perspective is useful for Cow Re-ID because the cow body and coat pattern are visible from the side. These visual features are important for identifying individual cows. However, the two farms differed in their environmental conditions and segmentation difficulty. Farm A had less severe metal-bar occlusion (Figure 3.2a), while Farm B had stronger vertical metal-bar occlusion in front of the cows (Figure 3.2b).



(a) Example side-view cow image from the Farm A dataset showing less severe metal-bar occlusion.



(b) Example side-view cow image from the Farm B dataset showing stronger vertical metal-bar occlusion.

Figure 3.2: Example side-view cow images from the two farm-specific datasets.

The data from the two farms were treated as separate datasets, each with its own set of cow identities. Therefore, the main experiments in this thesis are described as farm-specific experiments. Separate Re-ID models were trained for the Farm A and Farm B scenarios. For the final video-inference pipeline, the corresponding farm-specific DINOv3 model was used. This farm-specific training strategy was used because the cow identities, camera environment, occlusion level, and segmentation difficulty differed between the two farms. Table 3.1 compares the two farm-specific scenarios.

Table 3.1: Comparison of the two farm-specific dataset scenarios.

Aspect	Farm A	Farm B
Camera view	Side view	Side view
Main walking direction	Mainly left to right	Mainly right to left
Inference-gallery identities	Not used during training, validation, or testing	Present during training, validation, and testing
Occlusion level	Lower	Higher
Main occlusion source	Lower metal-bar occlusion	Stronger vertical metal-bar occlusion
Final video Re-ID model	Separate DINOv3 model trained on Farm A data	Separate DINOv3 model trained on Farm B data
Segmentation for inference	Default YOLO11s segmentation	Custom YOLO11m segmentation trained using SAM3-generated pseudo-labels
Re-ID image source	Multiple video recordings, dates, and daytime and night-time conditions	One 30-minute daytime video recorded on one date

3.2.1 Farm A Dataset

The Farm A dataset was used as one complete farm-specific Cow Re-ID scenario. The dataset contains 216 cow IDs and 7,462 images. The images were extracted from side-view videos and organised according to cow identity. Each cow image was assigned to the correct cow ID and stored in the corresponding cow folder.

The Farm A dataset was used to train, validate, and test the image-level Re-ID models for this farm. Since the split was performed at image level inside each cow ID folder, the same cow identities appeared in training, validation, and test sets, but with different images. This setup allowed the models to be evaluated on held-out images of known cow identities.

For final video inference, a separate setup was created. The 15 Farm A cow IDs used in video inference were not part of the training, validation, or test split. These cow IDs were enrolled only through separate gallery images during inference. The videos used for inference were also separate from the gallery images. Therefore, the gallery images were not created from the same videos used during video inference.

For the final Farm A video inference experiment, 15 cow IDs not used during DINOv3 training, validation, or testing were enrolled in the inference gallery. The gallery contained 311 original gallery images. After segmentation, 310 gallery images produced usable cow-only crops, while one image was skipped because no valid cow segmentation was produced.

The final inference was performed on 30 fixed one-minute videos, consisting of one daytime and one night-time target video for each of the 15 enrolled cow identities.

3.2.2 Farm B Dataset

The Farm B dataset was used as the second complete farm-specific Cow Re-ID scenario. The Re-ID image dataset was constructed from one 30-minute side-view video recorded on 2 October 2025 from 11:30 to 12:00. The video initially contained 193 RFID-identified cows.

The cow images were extracted from the video and organised into cow-ID folders. After image extraction, the images in every cow folder were inspected manually. Unusable, strongly blurred, incorrect, and visually redundant images were removed. Two cow IDs were excluded because their extracted images were too blurred. The final Farm B Re-ID dataset contained 191 cow IDs and 3,223 images.

The Farm B scenario was more challenging for segmentation because cows were often occluded by vertical metal bars in front of the camera. These metal bars covered parts of the cow body and made cow segmentation more difficult. Initial tests with the default YOLO11s segmentation model produced incomplete cow masks under these conditions. Therefore, a custom YOLO11m segmentation model trained using SAM3-generated pseudo-labels was used for gallery creation and video inference. A qualitative comparison between the default YOLO11s segmentation model and the custom YOLO11m segmentation model is presented in Section 6.3.1.

The custom YOLO11m segmentation model was trained using a separate pseudo-labelled dataset containing 3,681 images. The pseudo-label generation and custom segmentation training procedures are described in Sections 4.3.2.2 and 4.3.2.3.

Separate farm-specific Re-ID models were trained on the Farm B dataset containing 191 cow IDs. These models were used for the image-level model comparison, while the Farm B DINOv3 model was used in the final video-inference pipeline. Therefore, the Farm B experiment was treated as a separate farm-specific experiment.

To construct the final inference gallery, all 3,223 images from the 191 cow IDs were used. These images had been extracted from the video recorded on 2 October 2025. The gallery images were segmented using the custom YOLO11m segmentation model. All 3,223 images produced usable segmented cow crops, and no gallery image was skipped.

Final video inference was then performed using a separate 30-minute video recorded on 6 October 2025 from 11:30 to 12:00. This video was not used to create the inference gallery. The video contained 176 RFID-identified cows. Among them, 122 cow IDs were included in

the inference gallery, while 54 cow IDs were not included in the gallery. Therefore, the Farm B inference video contained both enrolled and non-gallery cows.

Together, the Farm A and Farm B datasets provide the image data used for farm-specific Re-ID model training and image-level evaluation. The query and gallery image counts in Table 3.2 refer only to the image-level retrieval evaluation and not to the final inference galleries. The main image dataset statistics for both farm scenarios are summarised in Table 3.2.

Table 3.2: Image dataset statistics for the two farm scenarios.

Dataset item	**Farm A**	**Farm B**
Cow IDs	216	191
Total images	7,462	3,223
Training images	5,217	2,178
Validation images	1,118	494
Test images	1,127	551
Image-level query images	396	247
Image-level gallery images	731	304

Unless stated otherwise, the experiments described in this thesis are farm-specific. For the image-level model comparison, each evaluated Re-ID architecture was trained and evaluated using data from the corresponding farm. For final video inference, the Farm A-trained DINOv3 model was used for the Farm A scenario, while the Farm B-trained DINOv3 model was used for the Farm B scenario. Additional cross-farm testing, in which the Farm A-trained DINOv3 model was applied to Farm B data, is presented in Section 6.5.4.

3.3 Preprocessing

This section describes the preprocessing steps used to prepare the cow image and video data for farm-specific Cow Re-ID. The preprocessing included image extraction from videos, identity-based dataset organisation, train/validation/test splitting, query–gallery preparation for image-level evaluation, gallery preparation for final inference, and segmentation-based crop extraction during video inference. These steps were applied separately to the Farm A and Farm B scenarios.

3.3.1 Image Extraction and Dataset Organisation

The image-extraction procedure differed between the two farms. For Farm A, cow images were manually selected and cropped from recorded videos. The videos were recorded on

different dates and under different daytime and night-time conditions. The images were then assigned to the correct cow IDs.

For Farm B, the images were extracted from one 30-minute video using a Python script. An RFID-based annotation file provided the cow ID and the first and last frames in which the full body of each cow was visible. The video had a frame rate of 25 frames per second. Images were extracted at an interval of 15 frames, corresponding to approximately one image every 0.6 seconds.

A farm-specific YOLO detector provided by Fraunhofer IGD was used to detect and crop the cow from each selected video frame. The cropped images were stored in the corresponding cow ID folders. The images in every folder were then inspected manually, and unusable, strongly blurred, incorrect, and visually redundant images were removed.

For both farms, the final images were organised using an identity-based folder structure, where each folder represented one cow ID and all images inside that folder belonged to the same cow. The general dataset structure is shown in Listing 3.1.

```
Cow_ReID_dataset/
    cow_ID_8816/
        image_001.jpg
        image_002.jpg
        image_003.jpg
    cow_ID_9230/
        image_001.jpg
        image_002.jpg
        image_003.jpg
```

Listing 3.1: Identity-based folder structure used for organising cow images by cow ID.

This structure is suitable for supervised Re-ID model training because each folder name represents one class label or cow identity. During training, the model learns to distinguish between cow IDs based on visual appearance. During retrieval evaluation, the same identity labels are used to check whether a query image is matched to the correct gallery identity.

YOLO segmentation was not used during image-level Re-ID model training in either farm scenario. The evaluated Re-ID models were trained using the prepared cow images and their cow ID labels. Segmentation was applied later during final inference gallery preparation and video inference, as described in Sections 3.3.4 and 3.3.6.

3.3.2 Train, Validation, and Test Split

For both farms, the train, validation, and test split was performed inside each cow ID folder. This means that the split was an image-level stratified split per cow ID, not an identity-level split. The same cow IDs appeared in the training, validation, and test sets, but with different images.

For each cow ID, the images were shuffled and split approximately according to a 70:15:15 ratio. For example, if one cow folder contained about 50 images, around 35 images were used for training, 7 or 8 images were used for validation, and 7 or 8 images were used for testing.

The final split ratios are not exactly 70:15:15 because the split was performed separately inside each cow-ID folder. Each cow had a different number of images, and integer rounding was required when assigning images to the training, validation, and test subsets. For cow IDs with at least four images, the splitting procedure also reserved at least two test images whenever necessary. This ensured that the same identity could contribute separate images to the query and image-level gallery sets. These requirements caused the final validation and test proportions to differ slightly from the target ratio. Table 3.3 shows the resulting split for both farms.

Table 3.3: Train, validation, and test split comparison.

Farm	Total images	Train	Validation	Test	Approximate ratio
Farm A	7,462	5,217	1,118	1,127	69.9 : 15.0 : 15.1
Farm B	3,223	2,178	494	551	67.6 : 15.3 : 17.1

Before splitting, the images in each cow-ID folder were inspected manually, and unusable, incorrect, strongly blurred, and visually redundant images were removed. Each image was assigned to only one of the training, validation, or test subsets. Therefore, the image-level evaluation used held-out images that were not used during model training.

For Farm A, the image dataset contained images collected from multiple video recordings, dates, and lighting conditions. For Farm B, the image-level split was created from selected images originating from one source video. Therefore, the Farm B image-level results represent closed-set retrieval within the same recording environment. Generalisation beyond the image-level split was evaluated separately through the final video-inference experiments: Farm A used separate target videos containing cow identities not used during model training, while Farm B used a separate video recorded on a different date.

3.3.3 Query–Gallery Split for Image-Level Evaluation

After the train, validation, and test split, query and gallery sets were created from the test split only. The test images were grouped by cow ID. For each cow ID, up to two test images were assigned to the query set. The remaining test images belonging to the same cow ID were assigned to the image-level gallery set.

YOLO segmentation was not applied to the query or gallery images used in this image-level retrieval evaluation. Instead, the prepared test images were passed directly through each evaluated farm-specific Re-ID model to extract embeddings. Each query embedding was then compared with the gallery embeddings using cosine similarity. The query–gallery matching concept was introduced in Section 2.7, and the retrieval metrics used in this evaluation, including Rank-k accuracy and mean average precision (mAP), were introduced in Section 2.10.

This was a closed-set image-level retrieval evaluation because the cow IDs in the query and gallery sets were already present during training. However, the evaluated query and gallery images were held-out test images. This image-level gallery was different from the final inference gallery described in Section 3.3.4. Table 3.4 shows the query–gallery split used for image-level retrieval evaluation.

Table 3.4: Query–gallery split for image-level evaluation.

Farm	Test images	Image-level query images	Image-level gallery images
Farm A	1,127	396	731
Farm B	551	247	304

3.3.4 Gallery Preparation and Segmentation Preprocessing

Gallery preparation was an important part of the final video-inference pipeline. The gallery-enrolment procedure differed slightly between the two farms. For Farm A, the cow IDs included in the final inference gallery were not part of the training, validation, or test sets. In contrast, for Farm B, the cow IDs included in the final inference gallery were also present in the training, validation, and test sets.

For both farms, the gallery images were separate from the videos used for inference. The gallery images were segmented to remove the surrounding background and retain the visible cow region. Each cow-only crop was then passed through the corresponding farm-specific DINOv3 model to create a gallery embedding. Figure 3.3 shows an example of this gallery

preprocessing step for both farm scenarios. The vector shown in Figure 3.3 is a shortened preview of the 256-dimensional Re-ID embedding produced by the farm-specific DINOv3 model, and only a few values are shown for readability.



Raw gallery image



Default YOLO11s segmented crop

```
[-0.004896, 0.023516,
 0.058115, 0.110972,
 0.077593, 0.007158,
-0.046880, ...]
```

DINOv3 embedding preview

(a) Farm A gallery preprocessing



Raw gallery image



Custom YOLO11m segmented crop

```
[-0.071625, -0.048513,
 0.029106, 0.041510,
 0.000743, 0.085696,
 0.047098, ...]
```

DINOv3 embedding preview

(b) Farm B gallery preprocessing

Figure 3.3: Example gallery preprocessing for Farm A and Farm B: raw gallery image, segmented cow crop, and shortened DINOv3 embedding preview.

The segmentation strategy differed between the two farms. For Farm A, the default YOLO11s segmentation model was used because the segmentation conditions were less challenging than in Farm B. For Farm B, initial tests showed that the default YOLO11s segmentation model produced incomplete cow masks under stronger vertical metal-bar occlusion. Therefore, a custom YOLO11m segmentation model trained using SAM3-generated pseudo-labels was used. A qualitative comparison between the default YOLO11s segmentation model and the custom YOLO11m segmentation model is presented in Section 6.3.1.

During gallery creation, one embedding was extracted from each usable gallery image. For each cow ID, the image-level embeddings were averaged and L2-normalised to form a single prototype embedding vector. During video inference, each segmented cow crop was converted into a query embedding and matched against these cow-level prototype embeddings using cosine similarity. Table 3.5 summarises the final inference gallery preparation for both farms.

Table 3.5: Final inference gallery preparation.

Farm	Cow IDs in inference gallery	Original gallery images	Usable segmented gallery images	Skipped gallery images	Segmentation method
Farm A	15	311	310	1	Default YOLO11s segmentation
Farm B	191	3,223	3,223	0	Custom YOLO11m segmentation trained using SAM3-generated pseudo-labels

3.3.5 Video Data Preparation for Final Inference

The final video inference setup was prepared separately from the image-level query–gallery evaluation. The inference videos were also separate from the images used to create the corresponding inference galleries.

For Farm A, final inference was performed on 30 fixed one-minute videos. Each of the 15 cows included in the inference gallery was evaluated in one daytime and one night-time target video. The 15 enrolled cow IDs were not used during DINOv3 training, validation, or image-level testing. The gallery images and inference videos were also obtained from different video recordings. Other cows could also appear in the inference videos.

For Farm B, final inference was performed on one continuous 30-minute video recorded on 6 October 2025 from 11:30 to 12:00. This video was separate from the 2 October 2025 video used to create the Re-ID dataset and inference gallery.

For both farms, the inference videos contained enrolled cows as well as cows that were not represented in the corresponding inference gallery. Because the system allowed an unknown decision for non-gallery cows, the final video experiments were open-set with respect to the inference gallery. Farm A additionally evaluated the enrolment of cow identities that had not been used during Re-ID model training, validation, or testing. Table 3.6 summarises the final video inference setup.

Table 3.6: Final video inference setup.

Farm	Number of videos	Video length	Video content	Video setting
Farm A	30	Fixed 1 minute each	One daytime and one night-time target video for each of the 15 enrolled cows; other cows could also appear in the videos.	Separate daytime and night-time side-view inference videos
Farm B	1	30 minutes	Continuous video containing enrolled and non-gallery cows	Independent continuous side-view inference video

The two farm scenarios evaluated different aspects of the proposed Cow Re-ID pipeline. The Farm A data were collected from different video recordings and dates, under both daytime and night-time conditions. The cow IDs included in the Farm A inference gallery were also

not used during model training. In contrast, the Farm B Re-ID dataset was constructed from one video recorded on a single date and time. Therefore, it contained less variation across different days and lighting conditions than the Farm A dataset. However, the Farm B final inference experiment used a larger gallery and one continuous video under stronger vertical metal-bar occlusion. Therefore, the two experiments should be interpreted as separate farm-specific evaluations rather than as a direct one-to-one comparison.

3.3.6 Preprocessing During Video Inference and Query Embedding Extraction

During video inference, a farm-specific detection, tracking, and segmentation procedure was used. For Farm A, the default YOLO11s segmentation model was used together with BoT-SORT tracking. YOLO11s produced the cow bounding box and segmentation mask, while BoT-SORT maintained the temporary track ID across consecutive video frames. For Farm B, the farm-specific YOLO detector first detected the cow, and BoT-SORT assigned the temporary track ID. The tracked cow crop was then segmented using the custom YOLO11m segmentation model. Figure 3.4 shows example video inference preprocessing results for both farm scenarios, including a raw video frame, the corresponding segmented cow crop, and a shortened preview of the DINOv3 query embedding.



Raw video frame



Default YOLO11s segmented crop

```
[0.119257, 0.031328,
-0.097505, 0.009163,
0.103086, 0.079684,
-0.053476, ...]
```

DINOv3 embedding preview

(a) Farm A video inference preprocessing



Raw video frame



Custom YOLO11m segmented crop

```
[0.046812, -0.082394,
0.117506, 0.028741,
-0.064923, 0.095318,
-0.013672, ...]
```

DINOv3 embedding preview

(b) Farm B video inference preprocessing

Figure 3.4: Example video inference preprocessing for Farm A and Farm B: raw video frame, segmented cow crop, and shortened DINOv3 query embedding preview.

After segmentation, each valid segmented cow crop was passed through the farm-specific DINOv3 model to obtain a query embedding. This query embedding represented the visual appearance of the detected cow in the current video frame. The query embedding was then compared with the cow-level prototype embeddings from the inference gallery using cosine similarity.

The video-inference pipeline also used tracking to maintain identity evidence across frames. Tracking was important because a cow could appear in multiple consecutive frames, and a single frame could be blurred, occluded, or poorly segmented. Therefore, the final identity decision was not based only on one isolated frame. Instead, the system used repeated embedding evidence across the video track to support a more stable identity decision.

If the similarity and accumulated track-level evidence satisfied the configured thresholds, the system displayed the matched cow ID. Otherwise, the track remained unknown. This unknown decision was important because the inference videos could contain non-gallery cows, and some tracked observations could also provide insufficient evidence because of blur, occlusion, or segmentation errors.

3.4 Summary

This chapter described the two farm-specific datasets and the preprocessing procedures used for image-level evaluation and final video inference. The Farm A dataset provided greater variation across recordings, dates, and lighting conditions, while the Farm B dataset provided a larger inference gallery under stronger metal-bar occlusion but with limited temporal variation. The next chapter presents the proposed farm-specific Cow Re-ID methodology, including Re-ID model training, segmentation, gallery prototype construction, tracking, and open-set identity matching.

4 Proposed Methodology for Farm-Specific Cow Re-ID

This chapter describes the methodology used to develop the farm-specific Cow Re-ID system. Chapter 2 reviewed the relevant literature, while Chapter 3 described the datasets and preprocessing steps. As introduced in Chapter 3, the two study sites are referred to as Farm A and Farm B.

The methodology has two connected stages. In the first stage, ResNet-50, MegaDescriptor, and DINOv3 were trained and evaluated separately for each farm using the farm-specific image datasets. Their performance was compared at image level, and DINOv3 was selected as the embedding model for the final video-inference pipeline. In the second stage, the selected farm-specific DINOv3 model was used in a gallery-based video-inference pipeline. In this pipeline, cows were localised, segmented, tracked, and compared with cow-level gallery prototypes.

Both farms followed the same general Re-ID principle. However, different localisation and segmentation methods were required because the farm conditions were different. Farm A used a pretrained YOLO11s segmentation model. Farm B used a farm-specific detector followed by a custom YOLO11m segmentation model trained using SAM3-generated pseudo-labels.

This chapter explains the main methodological steps and why they were used. Chapter 5 describes the software structure, exact attribute names, and file formats. Chapter 6 presents the experimental results.

Section 4.1 gives an overview of the complete methodology. Section 4.2 describes the DINOv3 architecture, training procedure, model selection, and image-level evaluation procedure. Section 4.3 presents the localisation and segmentation methods used for Farm A and Farm B. Section 4.4 explains how the cow-level gallery prototypes were created. Section 4.5 describes video-based Cow Re-ID, including tracking, query-embedding extraction, gallery matching, track-level evidence, identity locking, and the unknown decision. Finally, Section 4.6 describes the outputs produced by the methodology.

4.1 Overview of the Proposed Methodology

The proposed system follows a gallery-based Cow Re-ID approach. A reference gallery contains several images of each enrolled cow, while the inference video may contain both enrolled and non-enrolled animals. The objective is therefore not only to select the most similar gallery identity, but also to withhold an identity when the available evidence is weak or ambiguous. The complete workflow is illustrated in Figure 4.1.

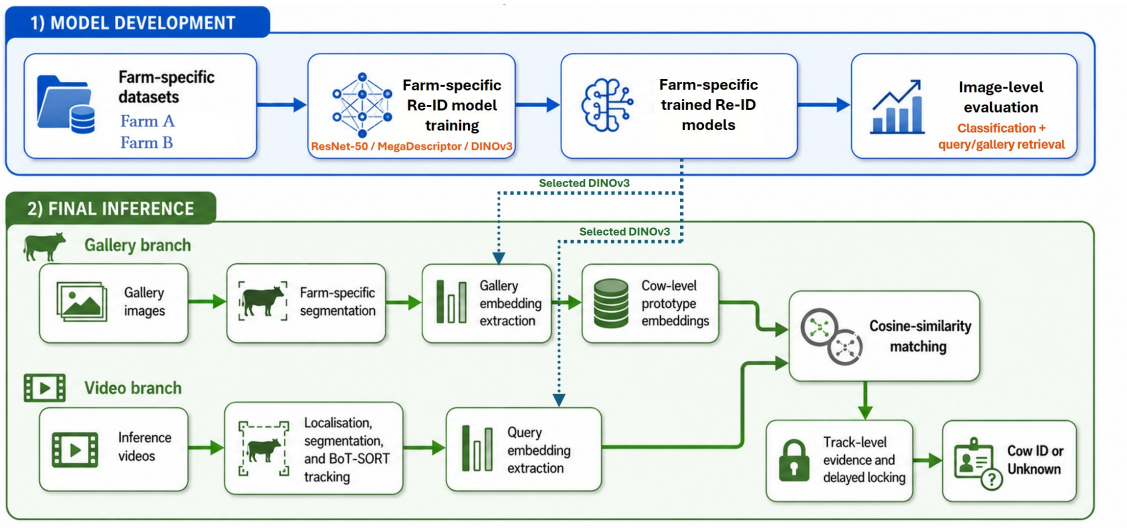


Figure 4.1: Overall methodology of the farm-specific Cow Re-ID system.

The methodology contains three main components:

1. farm-specific Re-ID model training and image-level evaluation;
2. farm-specific gallery preparation and cow-level prototype construction;
3. video-based identification using localisation, segmentation, BoT-SORT tracking, embedding matching, and track-level evidence.

During model training, the cow images described in Chapter 3 were used to train separate farm-specific ResNet-50, MegaDescriptor, and DINOv3 models for Farm A and Farm B. The models were trained using cow-ID classification to learn identity-discriminative visual representations. Each model produced a 256-dimensional embedding representation in addition to the cow-ID classification output. These embeddings were used for the image-level query–gallery evaluation. Based on the overall image-level comparison, DINOv3 was selected as the embedding model for the final video-inference pipeline, as discussed in Section 6.2.3.

After model training, the learned embeddings from ResNet-50, MegaDescriptor, and DINOv3 were evaluated using query and gallery images created from the held-out test split. This retrieval component of the image-level evaluation measured whether images of the

same cow were located close to one another in the embedding space and whether the correct gallery identity could be retrieved for each query image.

For final video inference, a separate inference gallery was constructed for each farm. In Farm A, the inference-gallery identities were not included during DINOv3 training, validation, or testing. In Farm B, the inference gallery contained the 191 identities that were represented during model development. Several reference images were available for each enrolled cow. Each valid gallery image was segmented and passed through the corresponding farm-specific DINOv3 model. The resulting image-level embeddings were averaged and L2-normalised to form one prototype embedding for each cow identity.

During video processing, cows were first localised and associated with temporary track IDs using BoT-SORT. The tracked cow regions were then segmented, and only observations with valid segmentation masks were eligible for query embedding extraction. Each query embedding was compared with all enrolled cow-level prototypes using cosine similarity. Frame-level matches that satisfied the similarity and ambiguity conditions contributed identity evidence to the corresponding track. A biological cow ID was assigned only after sufficient and consistent track-level evidence had been accumulated; otherwise, the track remained unknown.

Segmentation was not applied during image-level Re-ID model training. ResNet-50, MegaDescriptor, and DINOv3 were trained using the prepared cow images described in Chapter 3. Segmentation was introduced during final inference-gallery construction and video inference to reduce background information and to make the gallery and query inputs more comparable.

4.2 Farm-Specific DINOv3 Re-ID Model

This section describes the DINOv3-based Re-ID model used to learn visual representations of individual cows. The same architecture and training configuration were used separately for Farm A and Farm B. This produced one farm-specific checkpoint for each dataset. Only the dataset and the number of classification outputs were different.

ResNet-50 and MegaDescriptor were retained as comparison models. ResNet-50, MegaDescriptor, and DINOv3 were compared using the same image-level query–gallery evaluation protocol. MegaDescriptor was additionally integrated into the video-inference pipeline using the same localisation, segmentation, tracking, gallery-prototype construction, and track-level decision procedure as DINOv3. This enabled a video-level comparison between the selected DINOv3 model and an established embedding-based alternative. ResNet-50 was retained as a conventional image-level CNN baseline and was not integrated into the open-set

video-inference pipeline. The corresponding image-level and video-level comparisons are presented in Chapter 6.

Section 4.2.1 explains the network architecture and mathematical notation. Section 4.2.2 describes the training and model-selection procedure. Section 4.2.3 then explains the image-level evaluation procedure.

4.2.1 DINOv3 Network Architecture

DINOv3 was used as the main feature extractor in the Cow Re-ID pipeline. The implementation used the pretrained DINOv3 ViT-S/16 checkpoint released by the model authors [14]. A trainable embedding layer, a batch-normalisation layer, and a cow-ID classification layer were added to the pretrained backbone.

The notation used in this chapter is defined as follows. Scalar values and indices are written in italic type, bold lowercase symbols represent vectors and tensors, and bold uppercase symbols represent matrices. A circumflex represents an L_2-normalised quantity, while a tilde represents a temporally smoothed quantity. The superscript T represents transposition.

The input image tensor is written as $\mathbf{x}$, and $\boldsymbol{\theta}$ represents the trainable parameters of the DINOv3 backbone. Let $f_{\boldsymbol{\theta}}$ represent the DINOv3 backbone. The backbone feature vector is calculated as

$$\mathbf{h} = f_{\boldsymbol{\theta}}(\mathbf{x}), \tag{4.1}$$

where $\mathbf{h} \in \mathbb{R}^{d_h}$ is the pooled DINOv3 feature vector and d_h is the hidden size of the selected backbone.

A trainable linear layer converts the backbone feature into a 256-dimensional Re-ID embedding. Batch normalisation is then applied:

$$\mathbf{e} = \mathrm{BN}\left(\mathbf{W}_e\mathbf{h} + \mathbf{b}_e\right). \tag{4.2}$$

Here, $\mathbf{W}_e \in \mathbb{R}^{256\times d_h}$ is the embedding weight matrix, $\mathbf{b}_e \in \mathbb{R}^{256}$ is the embedding bias vector, and $\mathbf{e} \in \mathbb{R}^{256}$ is the resulting embedding vector.

During supervised training, the embedding is passed through the cow-ID classification layer:

$$\boldsymbol{\ell} = \mathbf{W}_c\mathbf{e} + \mathbf{b}_c, \tag{4.3}$$

where $\mathbf{W}_c \in \mathbb{R}^{C\times 256}$ is the classification weight matrix and $\mathbf{b}_c \in \mathbb{R}^{C}$ is the classification bias vector. The value C is the number of cow identities in the training dataset. Therefore, $C = 216$ for Farm A and $C = 191$ for Farm B. The output $\boldsymbol{\ell} \in \mathbb{R}^{C}$ is a vector of classification logits, with one score for each cow identity. It is not a matrix.

The network produces two outputs: the embedding vector $\mathbf{e}$ and the classification-logit vector ℓ. The classification logits are used during supervised training and closed-set classification evaluation. The embedding is used for image retrieval, gallery construction, and video inference. Before similarity is calculated, the embedding is L2-normalised:

$$\hat{\mathbf{e}} = \frac{\mathbf{e}}{\|\mathbf{e}\|_2}. \tag{4.4}$$

The circumflex in $\hat{\mathbf{e}}$ shows that the vector has unit Euclidean norm. This design allows the model to learn from labelled training identities and later use the learned embeddings for gallery matching. Figure 4.2 summarises how the pretrained DINOv3 backbone was adapted for supervised cow-ID training and embedding-based Re-ID inference.

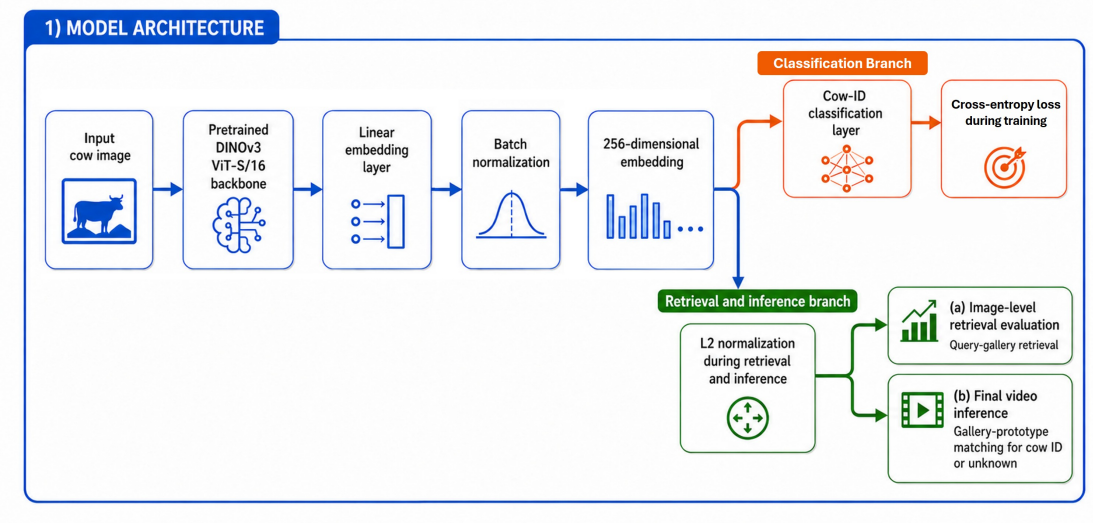


Figure 4.2: Adaptation of the pretrained DINOv3 backbone for supervised cow-ID training and embedding-based inference.

4.2.2 Training and Model Selection

The DINOv3 models for Farm A and Farm B were trained separately. Both models used the same architecture, optimisation method, augmentation, and hyperparameters. Only the farm-specific dataset and the number of cow-ID classes were different. As described in Section 3.3.2, the train–validation–test split was created inside each cow-ID folder. Therefore, the same cow identities appeared in all three subsets, but the images were different.

The input size and normalisation values associated with the pretrained DINOv3 checkpoint were used to resize and normalise the input images. During training, random horizontal flipping was applied with a probability of 0.5. This represented cows facing in opposite directions. Colour jitter varied brightness, contrast, and saturation by up to 0.2, while

hue was varied by up to 0.02. These moderate changes were used to improve robustness to orientation and illumination differences without strongly changing the coat patterns. Validation and test images were resized and normalised without random augmentation.

Cross-entropy loss was applied to the cow-ID logits during training. Let B represent the mini-batch size, $\ell_{i,k}$ the logit for image i and class k, and y_i the correct class. The average loss is

$$\mathcal{L}_{\mathrm{CE}} = -\frac{1}{B}\sum_{i=1}^{B}\log\left(\frac{\exp\left(\ell_{i,y_i}\right)}{\sum_{k=1}^{C}\exp\left(\ell_{i,k}\right)}\right). \tag{4.5}$$

The complete network was fine-tuned, including the pretrained backbone and the new embedding and classification layers. Both farm-specific models were trained for a maximum of 20 epochs with a batch size of 16. AdamW was used with a learning rate of 10^{-4} and a weight decay of 10^{-4}. Early stopping used a patience of five epochs and required a minimum validation accuracy improvement of 0.001. A new checkpoint was saved when the validation classification accuracy improved by more than 0.001 over the previously saved best value. Training stopped after five consecutive epochs without such an improvement, and the resulting best saved checkpoint was used for evaluation.

The values in Table 4.1 were chosen as practical fine-tuning settings for the available datasets and computing resources. They were not selected through an exhaustive hyperparameter search. Using the same configuration for both farms also allowed the two models to be compared without using different optimisation settings. Separate checkpoints were produced for Farm A and Farm B. The final image-level evaluation results are presented in Section 6.2.

Table 4.1: DINOv3 training configuration and selection rationale.

Parameter	Value	Rationale
Backbone and embedding	DINOv3 ViT-S/16; 256 dimensions	Uses pretrained visual features and produces a compact representation for prototype matching.
Optimiser	AdamW	Fine-tunes the transformer backbone together with the newly added layers.
Learning rate and weight decay	10^{-4}; 10^{-4}	Uses conservative updates and regularisation to reduce overfitting on the farm-specific datasets.
Batch size	16	Provides a practical balance between GPU-memory use and sample diversity.
Maximum epochs	20	Gives the model enough time to learn, while early stopping avoids unnecessary epochs.
Early stopping	Patience 5; minimum improvement 0.001	Allows short validation fluctuations but ignores very small changes.
Training augmentation	Flip $p = 0.5$; brightness, contrast, and saturation 0.2; hue 0.02	Adds moderate orientation and colour variation while preserving the coat pattern.
Model selection	Validation accuracy with minimum improvement of 0.001	Saves a new checkpoint when validation accuracy improves by more than 0.001.
Random seed	42	Supports reproducibility; the numerical value itself was arbitrary.

4.2.3 Image-Level Evaluation Procedure

After training, the selected checkpoint for each Re-ID model was evaluated at image level. Closed-set classification performance was evaluated using the held-out test images. In addition, query–gallery retrieval was performed using the query and gallery sets created from the held-out test split, as described in Section 3.3.3. For retrieval evaluation, each image was passed through the corresponding farm-specific Re-ID model, and the resulting 256-dimensional embedding was L2-normalised.

Let $\mathbf{E}_q \in \mathbb{R}^{N_q \times 256}$ and $\mathbf{E}_g \in \mathbb{R}^{N_g \times 256}$ represent the matrices of unnormalised query and gallery embeddings. Each embedding is stored as one row of the corresponding matrix. The row-wise L2-normalised matrices are written as $\hat{\mathbf{E}}_q$ and $\hat{\mathbf{E}}_g$. The circumflex shows that the

rows are normalised, while the bold uppercase notation shows that these quantities are matrices. The complete query–gallery similarity matrix is calculated as

$$\mathbf{S} = \hat{\mathbf{E}}_q \hat{\mathbf{E}}_g^T. \tag{4.6}$$

Each element S_{ij} is the cosine similarity between query image i and gallery image j. Because all rows are already L2-normalised, the matrix multiplication directly gives the pairwise cosine similarities.

Up to two held-out test images from each cow identity were used as query images. The remaining test images of the same identity were used in the image-level gallery. This was a closed-set image-level retrieval evaluation because the cow identities were represented during training, although the evaluated images were held out. Retrieval performance was measured using Rank-1 accuracy, Rank-5 accuracy, and mean average precision (mAP), as introduced in Section 2.10.

4.3 Farm-Specific Cow Localisation and Segmentation

Farm A and Farm B required different localisation and segmentation methods because their visual conditions were different. Segmentation was used to remove background content and keep the visible cow region before DINOv3 embedding extraction. The same farm-specific segmentation method was used for the corresponding inference gallery and video observations. Farm B also used additional quality checks during video inference.

4.3.1 Farm A Segmentation Pipeline

The Farm A pipeline used the pretrained `yolo11s-seg.pt` model [24, 25]. The COCO cow class, class ID 19, was selected with a confidence threshold of 0.30. During inference-gallery construction, YOLO11s segmentation was applied to every gallery image.

If an image contained more than one valid cow detection with a segmentation mask, the detection with the largest bounding-box area was selected. Each gallery image was intended to contain one main target cow. Therefore, smaller detections were more likely to represent a partially visible neighbouring cow or an incorrect detection. A padding ratio of 0.05 was added around the selected box, and all pixels outside the segmentation mask were set to zero.

Only gallery images with a valid cow segmentation mask were used for embedding extraction. Images without a valid mask were skipped and did not contribute to the cow-level prototype. The resulting gallery counts are reported in Section 3.3.4.

During video inference, YOLO11s segmentation and BoT-SORT tracking were used together. YOLO11s produced the current cow bounding box and segmentation mask. BoT-SORT assigned and maintained the temporary track ID across consecutive frames. The Re-ID crop was created from the current YOLO11s box and mask using the same 0.05 padding and zero-valued background used for the gallery.

4.3.2 Farm B Custom Segmentation Pipeline

Farm B contained stronger foreground occlusion from vertical metal bars than Farm A. Under these conditions, the default segmentation approach did not always preserve a sufficiently complete side-body region. Therefore, a custom cow-visible segmentation model was developed. Its purpose was to reduce the effect of the metal bars while keeping the coat regions needed for Re-ID.

Direct SAM3 segmentation was also tested as an alternative to the custom YOLO11m segmentation model. For direct inference, SAM3 used the text prompt "cow". This was different from the SAM3 pseudo-labelling procedure described in Section 4.3.2.2, which was used to create training annotations for the custom model. For the Farm B segmentation comparison, the downstream Re-ID configuration was kept consistent across the custom YOLO11m and direct SAM3 experiments. Both experiments used the same inference video, gallery identities and source gallery images, farm-specific DINOv3 checkpoint, detector, BoT-SORT configuration, crop and mask-quality criteria, similarity thresholds, and track-level identity-decision parameters. The gallery prototypes were rebuilt from the segmented gallery crops produced by the corresponding segmentation method. Therefore, the primary experimental difference was the segmentation approach used to generate the gallery and video cow crops. The qualitative segmentation comparison is presented in Section 6.3.2, while the quantitative video Re-ID comparison is presented in Section 6.5.2.

4.3.2.1 Farm-Specific Cow Detection and Tracking

A farm-specific YOLO detector supplied by Fraunhofer IGD was used to localise cows in Farm B. Its complete training procedure was outside the scope of this thesis, so it was used as an externally provided component. During video inference, the detector produced cow bounding boxes, and BoT-SORT connected these detections across frames to create temporary tracks. Crops produced from the detector boxes were also used as source images when developing the custom segmentation model.

4.3.2.2 SAM3-Based Pseudo-Label Generation

SAM3 was used to label the custom segmentation dataset automatically [26]. Each cow-centred crop received a box prompt that covered most of the image and one positive point near the centre. If SAM3 returned more than one mask, the largest mask was selected as the initial cow mask.

The masks were then cleaned to reduce narrow vertical-bar regions and small isolated regions, and to fill minor holes. The remaining cow contours were converted into normalised YOLO polygon annotations for the single class `cow_visible`. Because the labels were generated automatically, they were treated as pseudo-labels rather than manually verified ground truth. Random examples were inspected visually, but not every annotation was manually reviewed or corrected.

The pseudo-labelled dataset contained 3,681 images. It was divided into 2,945 training images and 736 validation images using a fixed random seed. The exact image-processing steps and implementation parameters are described in Chapter 5.

4.3.2.3 Custom YOLO11m Segmentation Training

A pretrained `yolo11m-seg.pt` model was fine-tuned using the generated polygon annotations. The model had one class and used an image size of 640 pixels, a batch size of 8, and a maximum of 100 epochs. Validation and mixed-precision training were enabled. Table 4.2 summarises the main training settings.

Table 4.2: Main configuration of the custom Farm B segmentation model.

Parameter	**Configuration**
Initial model	`yolo11m-seg.pt`
Segmentation class	`cow_visible`
Training/validation images	2,945 / 736
Image size	640
Maximum epochs	100
Batch size	8
Selected checkpoint	Best validation performance

The checkpoint with the best validation performance was used for Farm B gallery processing and video inference. Figure 4.3 shows the pseudo-label generation and custom-model training workflow. The qualitative Farm B segmentation comparisons are presented in Section 6.3.

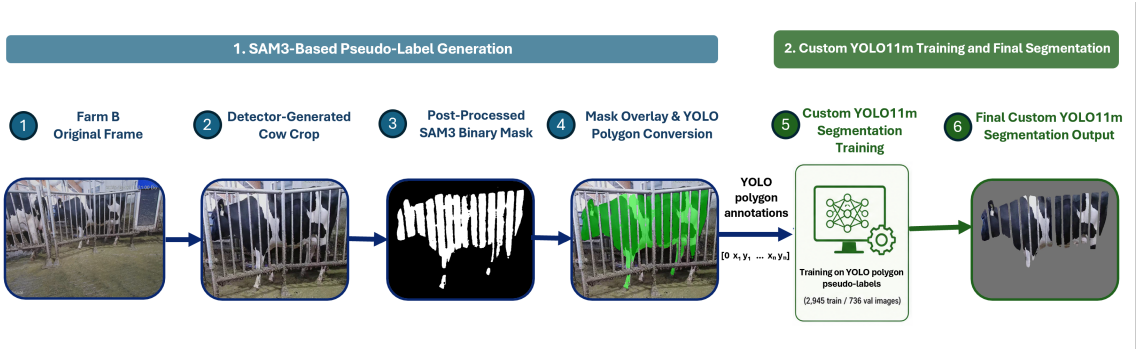


Figure 4.3: Workflow for SAM3-based pseudo-label generation and custom YOLO11m segmentation training for Farm B.

4.3.2.4 Final Farm B Gallery and Video Crop Extraction

The Farm B gallery images were already cow-centred detector crops. Therefore, the custom YOLO11m model was applied directly to these images without repeating cow detection. The predicted mask probabilities were thresholded at 0.40. One dilation operation with a 3×3 kernel was then applied to help preserve thin foreground regions and slightly expand the mask. Mask fragments covering at least 0.01, or 1%, of the crop area were retained and combined into a single visible-cow mask. Pixels outside the final mask were set to the neutral value 114. A DINOv3 embedding was created only when a valid cow segmentation mask was available. Gallery images without a valid mask were skipped and did not contribute to the cow-level prototype.

Video inference followed the same segmentation-only principle, but it also used additional quality checks. The first crop was extracted from the current raw detector box with a padding ratio of 0.15. If the boundary of a valid mask was within five pixels of the crop boundary, the same detector box was cropped again with a wider padding ratio of 0.25, and segmentation was repeated. A valid retry mask replaced the first mask. If the retry failed, the first valid mask was kept and passed to the quality gate. If the first segmentation did not produce a valid mask, the observation was skipped. Only observations with a valid segmentation mask were used for DINOv3 embedding extraction and gallery comparison.

A valid segmented video crop also had to contain a sufficiently complete horizontal cow-body region. Let $\mathbf{M}_t \in \{0,1\}^{H_t \times W_t}$ represent the final binary mask for observation t, where H_t and W_t are the height and width of the crop. The indices u and v refer to the row and column positions within the binary mask. A foreground cow pixel has the value 1, while a background pixel has the value 0.

Let $w_{M,t}$ and $h_{M,t}$ represent the width and height of the smallest bounding box enclosing the foreground mask, and let $w_{D,t}$ represent the width of the raw detector box. The mask-quality measures are defined as

$$r_{\text{area},t} = \frac{\sum_{u=1}^{H_t} \sum_{v=1}^{W_t} \mathbf{M}_t(u,v)}{H_t W_t}, \qquad r_{\text{width},t} = \frac{w_{M,t}}{w_{D,t}}, \qquad r_{\text{aspect},t} = \frac{w_{M,t}}{h_{M,t}}. \tag{4.7}$$

Here, $r_{\text{area},t}$ is the proportion of crop pixels belonging to the cow mask. The width ratio $r_{\text{width},t}$ measures how much of the detected cow width is covered by the segmented mask. The aspect ratio $r_{\text{aspect},t}$ measures whether the segmented region is wider than it is tall.

The crop was accepted for embedding extraction only when all three conditions were satisfied:

$$r_{\text{area},t} \geq 0.15, \qquad r_{\text{width},t} \geq 0.70, \qquad r_{\text{aspect},t} \geq 1.20. \tag{4.8}$$

The first condition required the cow mask to cover at least 15% of the crop. The second required the segmented region to cover at least 70% of the raw detector-box width. The third required the mask bounding box to be at least 1.20 times wider than it was tall. Together, these conditions rejected masks that were too small, too narrow relative to the detected cow, or consisted mainly of a vertical fragment. This prevented low-quality observations from entering the Re-ID evidence history.

4.4 Cow-Level Inference-Gallery Prototype Construction

The final inference gallery was different from the image-level gallery used in Section 4.2.3. The Farm A gallery contained 15 cow identities that were not used during DINOv3 training, validation, or testing. The Farm B gallery contained 191 cow identities that were also included in the image-level training and evaluation dataset. The complete gallery composition is given in Section 3.3.4.

The gallery images were stored in separate cow-ID folders and processed using the corresponding farm-specific segmentation method. Farm A used YOLO11s segmentation. Farm B applied the custom YOLO11m segmentation directly to the already-cropped gallery images. Only images with a valid segmented cow crop were passed through the corresponding DINOv3 model. Images without a valid segmentation mask were excluded and did not contribute to the prototype.

Assume that cow c has $N_c \geq 1$ usable gallery images. Let $\hat{\mathbf{e}}_{c,j} \in \mathbb{R}^{256}$ represent the L2-normalised embedding of gallery image j. The mean embedding is

$$\bar{\mathbf{p}}_c = \frac{1}{N_c} \sum_{j=1}^{N_c} \hat{\mathbf{e}}_{c,j}. \tag{4.9}$$

The mean embedding is then L2-normalised to create the final cow-level prototype:

$$\hat{\mathbf{p}}_c = \frac{\bar{\mathbf{p}}_c}{\|\bar{\mathbf{p}}_c\|_2}. \tag{4.10}$$

Here, $\hat{\mathbf{p}}_c$ denotes the 256-dimensional L2-normalised prototype of cow c. Thus, each enrolled cow is represented by a single unit-length prototype, regardless of the number of usable gallery images available for that cow.

Let M represent the number of enrolled cow identities. The cow-level prototypes are stacked row-wise to create the gallery-prototype matrix:

$$\hat{\mathbf{P}} = \begin{bmatrix} \hat{\mathbf{p}}_1^T \\ \hat{\mathbf{p}}_2^T \\ \vdots \\ \hat{\mathbf{p}}_M^T \end{bmatrix} \in \mathbb{R}^{M \times 256}. \tag{4.11}$$

Here, $\hat{\mathbf{P}}$ denotes the complete gallery-prototype matrix. Each row of $\hat{\mathbf{P}}$ contains the transposed 256-dimensional prototype of one of the M enrolled cow identities.

Averaging multiple usable gallery embeddings can reduce the influence of an unusual pose, image blur, partial occlusion, or an incomplete segmentation mask. The prototype matrix and cow-ID list were saved and reused for later videos instead of being rebuilt for every inference run. Figure 4.4 shows the complete prototype construction process.

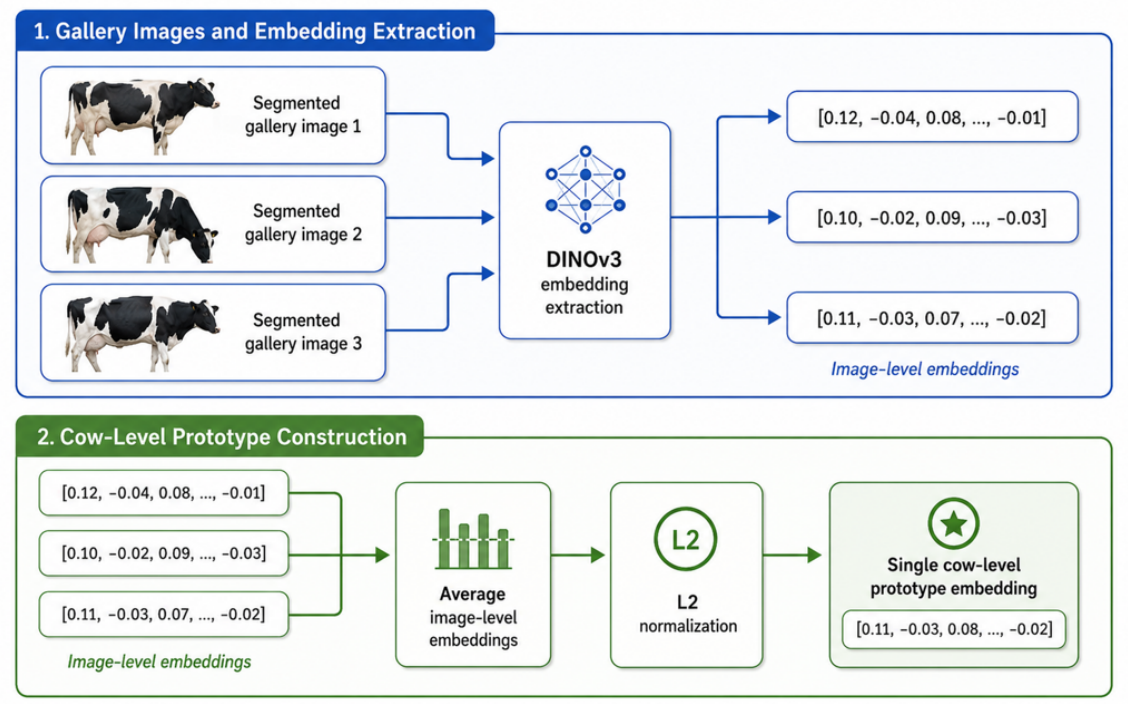


Figure 4.4: Construction of a cow-level prototype from multiple segmented gallery images of the same cow. The illustrated image-level vectors are L2-normalised and their numerical values are illustrative.

4.5 Video-Based Cow Re-Identification

The inference videos were processed frame by frame. The pipeline combined the farm-specific localisation and segmentation methods with BoT-SORT tracking, DINOv3 gallery matching, and delayed identity assignment. The following subsections explain temporary tracking, frame-level matching, track-level evidence, identity consistency, and the unknown decision.

4.5.1 Cow Localisation and BoT-SORT Tracking

For Farm A, the YOLO11s segmentation model produced the cow bounding box and segmentation mask. BoT-SORT [21] then assigned and maintained the temporary track ID across consecutive frames. For Farm B, the farm-specific detector produced the cow bounding box, BoT-SORT maintained the temporary track ID, and the custom YOLO11m model produced the segmentation mask inside the detector crop.

The temporary track ID was only an internal video identifier. It was not the biological cow ID. Its purpose was to connect observations of the same visible cow across several frames so

that Re-ID evidence could be collected over time. The biological cow ID was assigned only after gallery matching and track-level validation.

Bounding boxes can vary slightly between consecutive frames, which may produce visually unstable annotations. After localisation and track assignment, both scripts therefore applied exponential smoothing to the box used for display, reducing frame-to-frame jitter. This smoothing affected only output visualisation; segmentation, embedding extraction, and gallery matching used the current unsmoothed localisation and segmentation results.

Let $\mathbf{b}_t = [x_{1,t}, y_{1,t}, x_{2,t}, y_{2,t}]^T$ represent the current box used for display smoothing, and let $\tilde{\mathbf{b}}_t$ represent the smoothed box. The update is

$$\tilde{\mathbf{b}}_t = \alpha\tilde{\mathbf{b}}_{t-1} + (1-\alpha)\mathbf{b}_t, \qquad \tilde{\mathbf{b}}_1 = \mathbf{b}_1. \tag{4.12}$$

Here, t denotes the current video frame, while $\tilde{\mathbf{b}}_{t-1}$ is the smoothed bounding box from the previous frame. The coefficient $\alpha \in [0, 1]$ controls how strongly the previous smoothed box influences the current result. The term $1 - \alpha$ determines the contribution of the current box $\mathbf{b}_t$. A larger value of α produces smoother box movement but makes the smoothed box respond more slowly to rapid position changes. For the first observation of a track, no previous smoothed box exists; therefore, $\tilde{\mathbf{b}}_1$ is initialised using the first current box $\mathbf{b}_1$. The tilde indicates that the box has been temporally smoothed. The smoothing coefficient was 0.75 for Farm A and 0.85 for Farm B.

For Farm A, the Re-ID crop was obtained from the current YOLO11s segmentation result. For Farm B, the current raw detector box was used for cropping, followed by custom YOLO11m segmentation. For both farms, a track was finalised after it had not been observed for more than 40 frames, after which its active cow-ID ownership was released.

4.5.2 Query Embedding and Gallery Matching

After segmentation and the applicable farm-specific quality checks, the accepted cow crop was processed using the preprocessing configuration of the corresponding DINOv3 checkpoint. The resulting 256-dimensional query embedding was L2-normalised before being compared with the gallery prototypes.

Let $\hat{\mathbf{q}}_t$ represent the L2-normalised query embedding at frame t, and let $\hat{\mathbf{p}}_c$ represent the L2-normalised prototype of cow c. Their cosine similarity is calculated as

$$s_{t,c} = \frac{\hat{\mathbf{p}}_c^T \hat{\mathbf{q}}_t}{\|\hat{\mathbf{p}}_c\|_2 \|\hat{\mathbf{q}}_t\|_2}. \tag{4.13}$$

Since both vectors are L2-normalised, their norms are equal to one. Therefore, cosine similarity can be calculated using the dot product:

$$s_{t,c} = \hat{\mathbf{p}}_c^T \hat{\mathbf{q}}_t. \tag{4.14}$$

Since the prototypes of all enrolled cows are stored row-wise in $\hat{\mathbf{P}}$, the similarities between the query and all gallery prototypes can be calculated simultaneously:

$$\mathbf{s}_t = \hat{\mathbf{P}}\hat{\mathbf{q}}_t = \begin{bmatrix} \hat{\mathbf{p}}_1^T \hat{\mathbf{q}}_t \\ \hat{\mathbf{p}}_2^T \hat{\mathbf{q}}_t \\ \vdots \\ \hat{\mathbf{p}}_M^T \hat{\mathbf{q}}_t \end{bmatrix} \in \mathbb{R}^M. \tag{4.15}$$

Here, $\mathbf{s}_t$ is the gallery-similarity vector, and its c-th element, $s_{t,c}$, is the cosine-similarity score between the query at frame t and the prototype of cow c.

Let $s_{t,(1)}$ and $s_{t,(2)}$ represent the highest and second-highest values in $\mathbf{s}_t$, respectively. The frame-level similarity margin is defined as

$$m_t = s_{t,(1)} - s_{t,(2)}, \tag{4.16}$$

where m_t shows how clearly the best candidate is separated from the second-best candidate. A frame-level observation contributed identity evidence only when

$$s_{t,(1)} \geq \tau_s \quad \wedge \quad m_t \geq \tau_m, \tag{4.17}$$

where τ_s is the minimum similarity threshold and τ_m is the minimum margin threshold. The similarity threshold rejects weak matches, while the margin threshold rejects ambiguous matches.

The farm-specific threshold values were selected empirically through repeated inference runs and inspection of the matching behaviour on separate preliminary development videos. These development videos were not used for the final reported video-level evaluation. The parameters were not learned automatically and were not selected through an exhaustive search. Table 4.3 summarises the main differences between the two farms. Lower-level model and software settings are described in Chapter 5.

Table 4.3: Main farm-specific video-inference settings.

Method component	Farm A	Farm B
Localisation and segmentation	YOLO11s instance segmentation	Farm-specific detector followed by custom YOLO11m segmentation
Segmentation-mask refinement	Native YOLO11s instance mask	Threshold 0.40; one 3×3 dilation; merge valid fragments
Video crop padding	0.05	0.15; retry with 0.25 after boundary contact
Invalid video mask	Observation skipped	Observation skipped
Video mask-quality gate	Valid YOLO11s cow mask	Area ≥ 0.15, width ≥ 0.70, aspect ratio ≥ 1.20
Background outside mask	0	114
Similarity threshold τ_s	0.70	0.60
Frame margin threshold τ_m	0.10	0.05
Display-box smoothing	0.75	0.85

4.5.3 Track-Level Evidence Accumulation and Identity Locking

A biological cow ID was not assigned after only one accepted frame-level match. Instead, accepted evidence was collected over the corresponding track. This reduced the effect of a single blurred frame, partial occlusion, segmentation error, or temporary gallery confusion.

For each active track, the system stored a limited history of accepted candidate observations. For a candidate cow c, let n_c denote the number of accepted observations stored in the current evidence history. The index $r \in \{1, \ldots, n_c\}$ identifies one accepted observation, and $s_{c,r}$ denotes its cosine-similarity score with the prototype of cow c. The accumulated track-level score for candidate c is defined as

$$E_c = \sum_{r=1}^{n_c} s_{c,r} + 0.5 \max_{1 \leq r \leq n_c} s_{c,r} + 0.1 n_c. \tag{4.18}$$

Here, E_c is the accumulated evidence score for candidate cow c. The first term sums all accepted similarity scores in the current history. The second term adds half of the strongest accepted similarity score, giving additional weight to the best observation. The third term adds 0.1 for each accepted observation, thereby rewarding repeated support for the same candidate.

Let $E_{(1)}$ and $E_{(2)}$ denote the highest and second-highest track-level evidence scores among all candidate cow identities for the current track. The track-level score margin is defined as

$$M_{\text{track}} = E_{(1)} - E_{(2)}. \tag{4.19}$$

If only one candidate cow identity had accumulated evidence, the second-highest score $E_{(2)}$ was set to zero. Here, M_{track} measures how clearly the best-supported cow identity is separated from the second-best candidate. A larger value indicates a more confident track-level identity decision, while a small value indicates that the system is still uncertain between two competing cow identities.

Let c^* denote the cow identity with the highest accumulated track-level score. For this candidate, n_{c^*} is the number of accepted frame-level matches, and $\sum_{r=1}^{n_{c^*}} s_{c^*,r}$ is the sum of their accepted similarity scores. The candidate identity is locked to the current track only when all three conditions below are satisfied:

$$n_{c^*} \geq N_{\min}, \qquad \sum_{r=1}^{n_{c^*}} s_{c^*,r} \geq S_{\min}, \qquad M_{\text{track}} \geq M_{\min}. \tag{4.20}$$

Here, $N_{\min}$ is the minimum number of accepted frame-level matches, $S_{\min}$ is the minimum accumulated similarity required for the candidate, and $M_{\min}$ is the minimum required separation between the best and second-best track-level candidates. Therefore, an identity is locked only when it has been supported by enough accepted observations, has accumulated sufficient similarity evidence, and is clearly separated from its strongest competitor.

For both farms, evidence collection started once a track had been observed for at least five frames, and identity locking was not allowed before eight observed frames. The evidence history stored at most 35 accepted observations. The remaining farm-specific values are shown in Table 4.4. These values were selected empirically using the same separate preliminary development videos and were not learned automatically.

Table 4.4: Track-level identity-locking parameters.

Parameter	**Farm A**	**Farm B**
Start Re-ID after	5 track frames	5 track frames
Minimum observed frames before locking	8	8
Maximum evidence-history size	35	35
Minimum accepted matches $N_{\min}$	8	6
Minimum accumulated similarity $S_{\min}$	5.60	3.60
Minimum track-level margin $M_{\min}$	1.20	0.50
Track timeout	40 frames	40 frames

A track remained unknown until all locking conditions were satisfied. Figure 4.5 shows how the method moves from frame-level matching to delayed track-level identity locking.

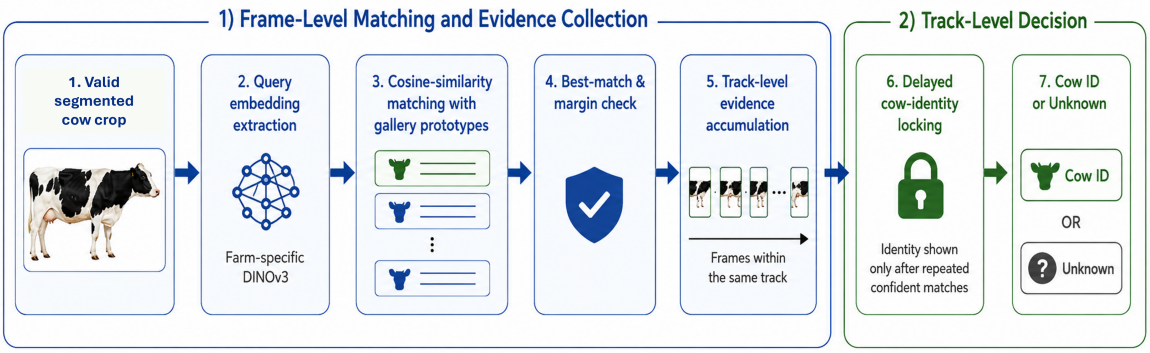


Figure 4.5: Track-level evidence accumulation and delayed cow-identity locking.

4.5.4 Identity Consistency, Unique-ID Control, and Unknown Decision

After an identity was locked, new valid query embeddings were still compared with the gallery. This consistency check was needed because a tracker could move from one cow to another during overlap, close interaction, occlusion, or temporary detector failure. Therefore, a locked identity was released only after repeated inconsistent evidence.

The reset rules were slightly different for the two farms. In Farm A, a low similarity to the locked identity alone did not release the ID. A different cow had to reach a similarity of at least 0.75 and exceed the locked prototype by at least 0.20. This had to happen in five consecutive consistency checks. The Farm A consistency threshold of 0.55 was used only to record the current observation as supporting evidence for the locked identity in the detected-ID statistics. It was not used as a separate reset condition.

In Farm B, the inconsistency counter increased when the similarity to the locked prototype was below 0.30. It also increased when a different best candidate had a similarity of at least 0.70 and a best-to-second-best margin of at least 0.12. The locked identity was released after 12 consecutive inconsistent checks.

When an identity was released, its active ownership was removed, the accepted evidence history was cleared, and the track returned to the evidence collection stage. Both pipelines also enforced unique ownership of cow IDs among active tracks. If one active track already owned a cow ID, another active track could not lock the same identity.

Table 4.5 summarises the final consistency and reset values. A track remained or temporarily became unknown when the evidence was weak, ambiguous, or insufficient, when a candidate identity was already owned by another active track, or when a previously locked identity was released after repeated inconsistency. After a reset, the track could collect new evidence and be identified again.

Table 4.5: Identity-consistency and reset parameters.

Parameter	Farm A	Farm B
Locked-ID consistency threshold	0.55, detected-ID statistics only	0.30, inconsistency condition
Minimum competing-ID similarity	0.75	0.70
Switching-separation criterion	Competitor minus locked ID ≥ 0.20	Best minus second-best ≥ 0.12
Consecutive inconsistent checks before reset	5	12
Evidence history cleared after reset	Yes	Yes
Re-identification after reset	Enabled	Enabled
Unique active ownership of cow IDs	Enforced	Enforced

An unknown decision did not always mean that localisation had failed. A cow could be detected, segmented, and tracked correctly but still remain unknown because the Re-ID evidence did not satisfy the acceptance and locking conditions. This prevented the system from forcing every visible cow into one of the enrolled gallery identities, which was important for the open-set inference videos.

4.6 Output Generation

The methodology produced annotated videos for visual inspection and structured records for later analysis. Only tracks that satisfied the track-level locking conditions received an enrolled cow-ID assignment. Tracks that did not satisfy these conditions remained unknown. For each locked track, the assigned cow ID and the associated similarity evidence were included in the output.

The structured records stored the information needed to inspect and evaluate each identity decision. Table 4.6 summarises the main output groups. The exact column names, data structures, and file formats are implementation details and are described in Chapter 5.

For Farm B, the first timestamp at which each cow ID was displayed and one corresponding crop were also stored. These outputs supported manual inspection and quantitative evaluation, but they did not change the identity decision. Chapter 5 describes how the outputs were stored, while Chapter 6 evaluates the Cow Re-ID results.

Table 4.6: Method-level information retained from video inference.

Information group	Purpose
Video and temporary track interval	Links each result to the processed video and the period during which the tracked cow was visible.
Final identity decision	Stores the locked cow ID and the frame at which the identity was assigned, while tracks that did not satisfy the locking requirements remain unknown.
Similarity and candidate evidence	Supports inspection of the winning identity, competing identities, and the strength of accumulated evidence.
Consistency and reset events	Records whether a locked identity remained stable or was released after repeated inconsistent observations.
Segmentation-quality diagnostics	Explains why Farm B observations were accepted, retried, or rejected before embedding extraction.
Identification timestamp and screenshot	Provides the first displayed Farm B identity time and one corresponding visual record for manual verification.

4.7 Summary

This chapter presented the proposed farm-specific Cow Re-ID methodology. ResNet-50, MegaDescriptor, and DINOv3 were trained and evaluated separately for each farm at image level, and DINOv3 was selected as the embedding model for the final video-inference pipeline. The selected farm-specific DINOv3 models were then used to create cow-level gallery prototypes. Farm A used a pretrained YOLO11s segmentation model. Farm B used a farm-specific detector, BoT-SORT tracking, and a custom YOLO11m segmentation model trained using SAM3-generated pseudo-labels. During inference-gallery construction and video inference, only valid segmented cow regions were used for embedding extraction. During video inference, query embeddings were compared with the gallery prototypes using cosine similarity. Track-level evidence was then collected before a cow identity was assigned or withheld. Chapter 5 describes the implementation of the methodology, while Chapter 6 presents the experimental evaluation.

5 Implementation and Experimental Setup

This chapter explains how the methodology from Chapter 4 was implemented and how the experiments were organised. It presents the software structure, computing environment, model training procedure, inference pipeline, saved outputs, and evaluation setup. Selected and simplified code excerpts are included to illustrate the main implementation decisions. Some surrounding loops, function arguments, validation steps, logging operations, and error handling are omitted for readability. The complete source code used in this thesis is available internally at Fraunhofer IGD. As the source-code repository is not publicly accessible, no repository URL is provided in this thesis. The numerical results and their interpretation are presented in Chapter 6.

Section 5.1 gives an overview of the software structure and the main project files. Section 5.2 describes the hardware and software environment used for the experiments. Section 5.3 explains the implementation of the Re-ID model training and image-level evaluation. Section 5.4 presents the SAM3 pseudo-label generation and custom YOLO11m segmentation training for Farm B. Section 5.5 describes the gallery construction and farm-specific video-inference pipelines. Section 5.6 explains the output files and stored attributes produced during inference. Finally, Section 5.7 presents the setup of the image-level, video-level, segmentation, embedding-model, and cross-farm experiments.

5.1 Implementation Overview

The Cow Re-ID system was implemented in Python. The same general training pipeline was used for ResNet-50, MegaDescriptor, and DINOv3. Separate inference scripts were used for Farm A and Farm B because the two farms used different localisation and segmentation procedures.

Table 5.1 summarises the main source files used in the project. The implementation was divided into reusable training modules, farm-specific segmentation code, and farm-specific video-inference code. This kept the common Re-ID operations separate from the parts that depended on the visual conditions of each farm.

Table 5.1: Main source files used in the Cow Re-ID implementation.

Source file	Main purpose
`common.py`	Dataset collection, per-identity image splitting, image transformations, data loading, embedding extraction, training, and retrieval metrics.
`models.py`	Construction of the ResNet-50, MegaDescriptor, and DINOv3 models with a common 256-dimensional embedding and cow-ID classification head.
`train_simple.py`	Shared training loop, early stopping, checkpoint selection, test classification, and image-level retrieval evaluation.
`run_resnet50.py`, `run_megadescriptor.py`, and `run_dinov3.py`	Model-specific entry points that call the shared training program with the selected backbone.
`generate_sam3_cow_visible_labels.py`	SAM3-based pseudo-label generation and conversion of cleaned masks to YOLO polygon annotations for Farm B.
`Farm_A_inference_script.py`	Farm A gallery construction and video inference using YOLO11s segmentation, BoT-SORT, and gallery-prototype matching.
`Farm_B_inference_script.py`	Farm B gallery construction and video inference using the Fraunhofer-provided detector, custom YOLO11m segmentation, BoT-SORT, mask-quality control, and structured output generation.
`evaluate_classification_extended.py`	Extended held-out test-set classification evaluation, including macro Precision, macro Recall, macro F1-score, per-class classification reports, and confusion matrices.

5.2 Hardware and Software Environment

The experiments were conducted on a Linux-based computing server running Ubuntu 22.04.5 LTS with Linux kernel 5.15.0. The server was equipped with an AMD EPYC 9534 processor, 491 GiB of system memory, and four NVIDIA RTX A6000 GPUs. Each GPU provided 49,140 MiB of memory, corresponding to approximately 48 GiB. One GPU was selected for each individual training or inference run. The installed NVIDIA driver version

was 580.173.02. The implementation used Python 3.10.12 and PyTorch 2.6.0. PyTorch used CUDA 12.4 with cuDNN 9.1.0. Table 5.2 lists the main software libraries used in the project.

Table 5.2: Main software environment used for implementation and experiments.

Software component	**Version**
Python	3.10.12
PyTorch	2.6.0
torchvision	0.21.0
CUDA runtime used by PyTorch	12.4
cuDNN	9.1.0
Transformers	5.8.0
Ultralytics	8.4.72
timm	1.0.26
OpenCV	4.13.0.92
scikit-learn	1.7.2
openpyxl	3.1.5
NVIDIA driver	580.173.02

5.3 Re-ID Model Training Implementation

This section describes how the Re-ID models were implemented, trained, and evaluated at image level. The same training pipeline was used for ResNet-50, MegaDescriptor, and DINOv3 to support a consistent comparison between the three models. The following subsections explain the dataset loading, model construction, training procedure, checkpoint selection, extended classification evaluation, and retrieval evaluation.

5.3.1 Dataset Loading and Split Generation

The training program expected one root directory containing one folder for each cow identity. All supported image files inside a cow folder received the same class label. Empty folders were skipped, and the remaining cow folders were sorted before numerical class indices were assigned.

The images of each cow identity were shuffled separately using random seed 42 and divided into training, validation, and test subsets. The target ratio was 70:15:15, as described in

Section 3.3.2. The split function also kept at least two test images for a cow whenever enough images were available. This was needed because the later retrieval evaluation required separate query and gallery images.

The held-out test subset was divided again for image-level retrieval. Up to two test images per cow were placed in the query set, while the remaining test images of the same cow were placed in the image-level gallery. The same image was never used in both sets. Identities with fewer than two test images would be excluded from retrieval evaluation, although the final datasets produced valid query and gallery sets for both farms.

Training images were shuffled by the PyTorch data loader. Validation and test images were loaded without shuffling. Pinned memory was enabled when a CUDA device was available.

5.3.2 Model Construction and Image Processing

The three Re-ID backbones were constructed through one model factory. Each backbone was connected to the same 256-dimensional embedding head and cow-ID classification layer described in Chapter 4. This common output structure allowed the same training and evaluation code to be used for all three models.

ResNet-50 was initialised with the pretrained torchvision weights. MegaDescriptor used the pretrained `BVRA/MegaDescriptor-T-224` model through the `timm` library. DINOv3 used the pretrained `facebook/dinov3-vits16-pretrain-lvd1689m` checkpoint through the Transformers library. ResNet-50 and MegaDescriptor used an input size of 224×224 pixels. For DINOv3, the input size and normalisation values were read from the image processor stored with the pretrained checkpoint.

The shared training transformation resized the image, applied random horizontal flipping with probability 0.5, and used colour jitter with brightness, contrast, and saturation values of 0.2 and a hue value of 0.02. Validation, test, gallery, and query images were resized and normalised without random augmentation.

5.3.3 Training, Early Stopping, and Checkpoint Storage

The shared training program used cross-entropy loss and the AdamW optimiser. All parameters of the pretrained backbone and the added layers were updated. The maximum number of epochs was 20, the batch size was 16, the learning rate was 10^{-4}, and the weight decay was 10^{-4}. The final Re-ID training runs used four data-loading workers. Early stopping used a patience of five epochs. A validation-accuracy improvement had to exceed 0.001 before the patience counter was reset.

After every epoch, the program stored the training loss, training classification accuracy, and validation classification accuracy. When the validation accuracy improved by more than the configured minimum threshold of 0.001, the checkpoint was overwritten with the current model. The selected checkpoint file, `best_model.pt`, contained the model's name, model state, cow-ID class mapping, and image-preprocessing configuration. This allowed the same file to be loaded later as an embedding extractor during gallery construction and video inference. Listing 5.1 shows the validation-based checkpoint selection and early-stopping logic used during model training.

```
if val_acc > best_val + args.min_delta:
    best_val = val_acc
    epochs_without_improvement = 0

    torch.save(
        {
            "model_name": args.model,
            "state_dict": model.state_dict(),
            "class_to_idx": class_to_idx,
            "prep": prep,
        },
        outdir / "best_model.pt",
    )
    print("Saved new best model.")

else:
    epochs_without_improvement += 1
    print(
        f"No improvement for {epochs_without_improvement} epoch(s). "
        f"Patience = {args.patience}"
    )

    if epochs_without_improvement >= args.patience:
        print(f"Early stopping triggered at epoch {epoch}.")
        break
```

Listing 5.1: Validation-based checkpoint selection and early stopping.

The training program created the following output files for each model and farm:

- `best_model.pt`: selected model checkpoint;
- `history.json`: epoch-wise training and validation history;
- `results.json`: final classification and retrieval measures;
- `cmc_curve.png`: CMC curve for the image-level retrieval evaluation.

5.3.4 Extended Classification Evaluation

In addition to the classification accuracy produced by the shared training program, an extended classification-evaluation script was used to calculate macro Precision, macro Recall, and macro F1-score on the held-out test set. The script reconstructed the original train–validation–test split using the same random seed and split ratios as model training and verified that the dataset class mapping matched the selected checkpoint.

The selected checkpoint was loaded using its saved image-preprocessing configuration, and predictions were obtained from the cow-ID classification output for all held-out test images. Macro Precision, macro Recall, and macro F1-score were then calculated across the cow identities represented in the test set.

The script also generated per-class classification reports, per-image predictions, and count and normalised confusion matrices for diagnostic inspection. However, because the classification tasks contained 216 cow identities for Farm A and 191 for Farm B, the confusion matrices were visually dense and difficult to interpret at the individual class level. Therefore, they were not included in the final experimental results. The reported classification comparison used classification accuracy, macro Precision, macro Recall, and macro F1-score.

5.3.5 Image-Level Retrieval Implementation

After training, the selected checkpoint was loaded and used to extract L2-normalised embeddings from the query and gallery images. The complete similarity matrix was calculated through matrix multiplication, following Equation 4.6. Gallery images were sorted by decreasing cosine similarity for every query.

The retrieval implementation calculated Rank-1 accuracy, Rank-5 accuracy, mean average precision, and the first 20 values of the CMC curve. The CMC values and curve were retained as auxiliary implementation outputs and were not included in the final reported model comparison, which used Rank-1, Rank-5, and mAP as the retrieval measures. Average precision was calculated only for queries whose cow identity was present in the gallery. The final query and gallery sizes and the number of valid queries were also stored in `results.json`. The image-level results of the three models are presented in Section 6.2.

5.4 Farm B Segmentation Implementation

This section explains the implementation of the segmentation approach used for Farm B. SAM3 was first used to generate pseudo-labels from the prepared cow images, and these

labels were then used to train a custom YOLO11m segmentation model. The implementation steps for pseudo-label generation, model training, and qualitative mask comparison are described in the following subsections.

5.4.1 SAM3 Pseudo-Label Generation

The Farm B segmentation dataset was created from cow-centred image crops. The pseudo-label generator processed these images with SAM3 and saved the source image, binary mask, visual overlay, and YOLO polygon label in separate training and validation directories.

For each image, the SAM3 box prompt covered the cow crop except for a 5% border on each side. One positive point was placed at the centre of the crop. When more than one mask was returned, the mask with the largest area was selected. The selected mask was cleaned before conversion to a YOLO polygon. The cleaning procedure removed detected narrow vertical structures, applied one 3×3 morphological opening, and then applied one 3×3 morphological closing.

Table 5.3 lists the main implementation settings. Masks outside the accepted area range were skipped. Connected components and contours smaller than 100 pixels were removed. The remaining contours were simplified and converted to normalised YOLO polygon coordinates.

Table 5.3: Main SAM3 pseudo-label generation settings.

Parameter	**Implementation value**
Segmentation class	`cow_visible`, class ID 0
SAM3 image size	518 pixels
Box prompt	Cow crop excluding a 5% border on each side
Positive point	Centre of the crop
Initial mask threshold	0.50
Accepted mask-area ratio	0.03 to 0.95
Minimum connected-component area	100 pixels
Polygon approximation ratio	0.003 of contour perimeter
Vertical-bar removal	Enabled
Training/validation split	80% / 20%
Random seed	42

The vertical-bar removal step was implemented as a heuristic. Each image was converted to grayscale, smoothed with a 3×3 Gaussian filter, and processed with Canny thresholds 50 and 150. A vertical morphological-opening kernel of width 3 and height $\max(20, \lfloor h/3 \rfloor)$ was used, where h is the image height. The detected vertical regions were dilated once with a 5×5 kernel. A region was treated as a possible bar when its height exceeded 35% of the image height and its width was below 15% of the image width. A small padding of at least two pixels was added before the region was removed from the initial SAM3 mask.

The generated labels were pseudo-labels and not manually drawn pixel-level ground truth. The saved masks and overlays allowed visual inspection. A subset of generated examples was inspected manually, but the complete pseudo-labelled dataset was not manually corrected.

5.4.2 Custom YOLO11m Segmentation Training

The cleaned YOLO polygon labels were used to fine-tune `yolo11m-seg.pt`. The training was launched through Ultralytics with one segmentation class. The final run used the settings listed in Table 5.4. Mixed-precision training and validation were enabled. Ultralytics' automatic optimiser selection and the augmentation configuration recorded in `args.yaml` were retained.

Table 5.4: Implementation settings of the final custom YOLO11m segmentation run.

Parameter	**Value**
Initial checkpoint	`yolo11m-seg.pt`
Number of classes	1
Maximum epochs	100
Early-stopping patience	100
Batch size	8
Image size	640 pixels
Data-loading workers	8
GPU selection	Device 0
Mixed precision	Enabled
Validation during training	Enabled
Deterministic execution	Enabled
Ultralytics random seed	0
Selected checkpoint	`best.pt`

The training run stored its parameter file, epoch-wise metric file, and training plots as `args.yaml`, `results.csv`, and `results.png`. The checkpoint with the best validation performance was used in the final Farm B gallery and video pipeline.

5.4.3 Qualitative Segmentation Comparison

Three segmentation approaches were compared visually on Farm B images: default YOLO11s segmentation, direct SAM3 segmentation, and the custom YOLO11m segmentation model. The comparison was qualitative because no manually annotated pixel-level ground-truth masks were available for calculating segmentation metrics such as Intersection over Union (IoU) or the Dice coefficient.

The produced segmented cow crops were inspected manually. The inspection considered whether the visible cow body was retained, whether important body regions were missing, how much background remained, whether vertical bars or nearby cows were included, and whether the final crop was suitable for Re-ID embedding extraction. This comparison was used as an initial segmentation model-selection step. The qualitative segmentation examples and findings are presented in Section 6.3.

5.5 Inference-Gallery and Video-Pipeline Implementation

This section describes how the trained Re-ID models were used during gallery construction and video inference. It explains checkpoint loading, cow-level prototype generation, gallery caching, farm-specific video processing, and track-level identity management. The implementation differs between the two farms mainly in the localisation and segmentation stages, while the general gallery-matching procedure remains the same.

5.5.1 Checkpoint Loading and Gallery Prototype Cache

During inference, the selected Re-ID checkpoint was loaded first. The model type, class mapping, trained model weights, and preprocessing configuration were read from `best_model.pt`. Although the classification layer remained part of the loaded network, only the 256-dimensional embedding output was used for gallery matching.

The gallery images were collected from cow-ID folders and processed using the farm-specific segmentation method. Every accepted image embedding was L2-normalised. The embeddings belonging to the same cow identity were averaged and normalised again, as

defined in Equations 4.9 and 4.10. The resulting cow-level prototype vectors were stacked into the prototype matrix defined in Equation 4.11.

To avoid rebuilding the gallery for every inference run, the cow-ID list and prototype matrix were stored in a gallery-cache file. When a gallery-cache file was available and gallery rebuilding was not requested, the stored prototypes were loaded directly. Otherwise, the gallery embeddings and prototypes were recomputed and the resulting cache was saved for later inference runs. Listing 5.2 shows a simplified excerpt of the cow-level gallery-prototype construction.

```
embs = torch.stack(cow_to_embs[cow_id], dim=0)
proto = F.normalize(embs.mean(dim=0),dim=0)
prototype_ids.append(cow_id)
prototype_vecs.append(proto)
prototype_matrix = torch.stack(prototype_vecs,dim=0).to(device)
```

Listing 5.2: Simplified cow-level gallery-prototype construction.

The cache also stored metadata describing how the gallery was created. Depending on the farm, this included the gallery root, Re-ID checkpoint, segmentation model or settings, preprocessing information, and background value.

5.5.2 Farm A Inference Implementation

Farm A used the pretrained YOLO11s segmentation model for both gallery processing and video inference. The COCO cow class, class ID 19, was selected with a confidence threshold of 0.30. For each gallery image, the valid cow detection with the largest bounding-box area was selected. A padding ratio of 0.05 was added, and pixels outside the mask were set to zero. Full-image fallback was disabled; therefore, an image without a valid cow mask was skipped.

For video inference, YOLO11s and BoT-SORT were called through the Ultralytics tracking interface with persistent tracks. Each current segmentation mask was resized to the video-frame size. The resized mask and video frame were then cropped using the padded cow bounding box, and the resulting mask was applied to the crop. After a valid segmented crop was obtained, it was passed to the Re-ID model and processed using the common query-embedding and gallery-matching procedure described in Section 5.5.4.

Annotated videos were written using the MP4V codec. A green rectangle and the locked cow ID were drawn only after the identity-locking conditions had been satisfied. The display box was smoothed with coefficient 0.75, while the current-frame YOLO11s segmentation result was used to create the Re-ID crop. The active values for frame-level matching, track-

level locking, identity consistency, timeout, and display smoothing were those reported in Tables 4.3, 4.4, and 4.5.

5.5.3 Farm B Inference Implementation

Farm B used the detector checkpoint supplied by Fraunhofer IGD. The detector was integrated as an external component and was not trained in this thesis. Its cow class was class ID 2. The active detector confidence threshold was 0.25 and its inference image size was 1280 pixels. BoT-SORT was used to maintain temporary track IDs.

The custom YOLO11m segmentation model was applied inside each detector crop. The segmentation class was class ID 0, the model confidence threshold was 0.20, and the inference image size was 1024 pixels. The predicted masks were thresholded at 0.40, dilated once with a 3×3 kernel, and all valid mask fragments were combined into a single visible-cow mask. A fragment had to cover at least 1% of the crop area. Pixels outside the final mask were set to 114. No bounding-box fallback was used.

The first video crop used 0.15 padding. When the boundary of a valid mask lay within five pixels of the crop boundary, the same raw detector box was cropped again with 0.25 padding. If the wider retry produced a valid mask, it replaced the first mask. If the retry failed, the first valid mask was retained and passed to the quality gate. The area, width, and aspect-ratio conditions from Equation 4.8 were then applied. Only segmented crops that passed these quality checks were used for query-embedding extraction and gallery matching. This common procedure for both farms is described in Section 5.5.4.

A strict segmentation-only rule was applied during Farm B video inference. When the segmentation model did not return a valid cow-visible mask, or when the mask failed the quality checks, the current observation was skipped. Therefore, no query embedding or gallery comparison was produced from an invalid or low-quality segmented crop. The simplified implementation is shown in Listing 5.3.

```
crop_bgr, seg_mask, seg_source = \
    apply_yolo_segmentation_to_crop(...)

if (
    crop_bgr is None
    or seg_mask is None
    or seg_source != "cow_visible_seg"
):
    state["no_crop_frames"] += 1
    continue

quality = evaluate_reid_mask_quality(
```

```
    seg_mask,
    min_area_ratio=reid_min_mask_area_ratio,
    min_width_ratio=reid_min_mask_width_ratio,
    min_aspect_ratio=reid_min_mask_aspect_ratio,
    reference_width=raw_detection_width,
)

if not quality["accepted"]:
    state["quality_rejected_frames"] += 1
    continue
```

Listing 5.3: Simplified rejection of invalid or low-quality segmentation results before Re-ID.

Re-ID processing started once a track had been observed for at least five frames, and identity locking was not permitted before eight observed frames. The active history size, matching thresholds, locking conditions, identity-reset values, and timeout were the Farm B values given in Tables 4.3, 4.4, and 4.5. The smoothed box with coefficient 0.85 was used only for drawing and for the saved identification screenshot.

5.5.4 Query Embedding and Gallery Matching

After a valid segmented cow crop was obtained, the crop was passed through the selected Re-ID model to generate a 256-dimensional L2-normalised query embedding. This procedure was the same for Farm A and Farm B.

The query embedding was compared with all cow-level gallery prototypes using the gallery-similarity operation defined in Equation 4.15. Because the query embedding and gallery prototypes were L2-normalised, the comparison was equivalent to the cosine-similarity calculation in Equation 4.14. The highest and second-highest similarity scores were retained together with their corresponding cow IDs. Their difference was then calculated according to Equation 4.16 and used as the frame-level similarity margin. The accepted frame-level match information was then passed to the track-level evidence procedure described in the following subsection. Listing 5.4 shows a simplified excerpt of the query-to-gallery matching procedure used in both farm-specific inference pipelines.

```
emb = extract_embedding(reid_model, transform, crop_bgr, device)
sims = torch.matmul(emb, prototype_matrix.T).squeeze(0)
topk = torch.topk(sims, k=min(2, sims.numel()))

best_idx = topk.indices[0].item()
best_sim = topk.values[0].item()
best_id = prototype_ids[best_idx]

if sims.numel() >= 2:
```

```
    second_idx = topk.indices[1].item()
    second_sim = topk.values[1].item()
    second_id = prototype_ids[second_idx]
else:
    second_id = None
    second_sim = -1.0

margin = best_sim - second_sim
```

Listing 5.4: Simplified query-to-gallery prototype matching.

5.5.5 Track State and Identity Management

Each active track was represented by a dictionary including its first and last frame, number of observed frames, bounded evidence history, most recent best gallery candidate, similarity margin, locked identity, identity-reset counter, and smoothed display box. Farm B additionally stored the most recent second-best gallery candidate together with segmentation and quality-control counters.

Accepted frame-level observations were stored in a double-ended queue with a maximum length of 35. The evidence was summarised by cow ID and converted to the track-level score in Equation 4.18. The software then checked the count, accumulated similarity, and track-margin conditions from Equation 4.20. Active cow-ID ownership was stored in a separate dictionary so that two active tracks could not hold the same biological cow ID at the same time. Listing 5.5 shows the shared implementation steps used to rank the accumulated candidate evidence and check the three track-level identity-locking conditions.

```
# Summarise accepted evidence for each candidate cow ID.
ranked = []

for cow_id, s in stats.items():
    final_score = (
        s["sum_sim"]
        + 0.5 * s["max_sim"]
        + 0.1 * s["count"]
    )

    ranked.append({
        "cow_id": cow_id,
        "count": s["count"],
        "sum_sim": s["sum_sim"],
        "max_sim": s["max_sim"],
        "final_score": final_score,
    })

ranked = sorted(
```

```
    ranked,
    key=lambda x: x["final_score"],
    reverse=True,
)

if ranked:
    best = ranked[0]
    second_score = (
        ranked[1]["final_score"]
        if len(ranked) > 1
        else 0.0
    )

    score_margin = best["final_score"] - second_score

    if (
        best["count"] >= min_lock_hits
        and best["sum_sim"] >= min_total_similarity
        and score_margin >= track_min_margin
    ):
        # The candidate satisfies the track-level
        # identity-locking conditions.
        ...
```

Listing 5.5: Simplified track-level evidence scoring and lock-condition evaluation.

When a track had not been observed for more than 40 frames, it was finalised and removed from the active state. Its cow-ID ownership was also released. After an identity reset, the track returned to evidence collection and could later receive another identity.

5.6 Output Files and Stored Attributes

Both inference pipelines produced annotated videos and track-level CSV files. The saved attributes made it possible to inspect how an identity decision was reached instead of retaining only the final displayed cow ID.

For Farm A, the main structured file was `track_level_manual_inspection.csv`. Each row represented one completed temporary track and included the video name, track ID, first and last frame, number of observed frames, final predicted ID, lock frame, similarity evidence, identity-reset count, candidate statistics, and empty manual-review columns. The manual columns allowed the predicted identity, true identity, and reviewer note to be added later. For Farm B, the main track-level file was `track_level_results_seg_quality_soft.csv`. In addition to the common track and identity fields, it stored:

- numbers of valid and rejected segmentation frames;
- numbers of embeddings created;
- adaptive edge-retry attempts and successes;

- mask-quality rejection reasons;
- displayed-ID, detected-ID, and candidate-count statistics.

Farm B also created one timestamp record when a cow ID was displayed for the first time in a video. The per-video CSV used the columns `Cow ID`, `Timestamp`, and `Screenshot`. The same information was written to an Excel file with the screenshot embedded in the third column. Screenshots were stored in a separate folder and linked to the timestamp records. Debug crops could also be saved every 50 frames to show the exact segmented input passed to the Re-ID model. These outputs supported the RFID-based evaluation described in Section 5.7.6.

5.7 Experimental Setup

This section presents the setup used to evaluate the implemented Cow Re-ID pipeline. The experiments include image-level comparison of the three Re-ID models, qualitative comparison of the Farm B segmentation methods, video-level Re-ID experiments, comparison between DINOv3 and MegaDescriptor, and cross-farm transfer from Farm A to Farm B. It also explains the ground-truth verification procedure and the measures used to evaluate the video-inference results. The inference thresholds and track-level decision parameters were selected using separate preliminary development videos before the final evaluation experiments were conducted.

5.7.1 Image-Level Re-ID Model Comparison

ResNet-50, MegaDescriptor, and DINOv3 were trained and evaluated separately on Farm A and Farm B. All three models used the same per-identity train–validation–test split, random seed, 256-dimensional embedding size, augmentation, optimiser, learning rate, early-stopping rule, and image-level evaluation code. Each model used its own pretrained backbone and preprocessing values.

The comparison used the held-out query and gallery sets described in Section 3.3.3. The reported measures were test classification accuracy, macro precision, macro recall, macro F1-score, Rank-1, Rank-5, and mAP. This experiment examined how the three feature extractors performed under the same farm-specific closed-set image-level protocol.

5.7.2 Final Farm-Specific DINOv3 Video Experiments

The final Farm A experiment used 30 fixed one-minute videos and an inference gallery containing 15 cow IDs. Each enrolled cow in the inference gallery was evaluated in one daytime and one night-time target video. These 15 identities were unseen during Farm A model training, validation, and image-level testing. The gallery images and evaluation videos were obtained from separate video recordings. Other cows that were not enrolled in the gallery could also appear in the videos, making the inference setting open-set.

The final Farm B experiment used one continuous 30-minute video and an inference gallery containing 191 cow IDs. According to the RFID records, the video contained 176 cow identities. Of these, 122 were represented in the gallery and 54 were not represented. The Farm B video therefore also formed an open-set evaluation. The gallery and inference-video composition are described in Sections 3.2.1, 3.2.2, and 3.3.5.

The separate farm-specific DINOv3 checkpoint, segmentation procedure, and inference settings were used for each farm. Because the farms differed in dataset composition, gallery size, recording conditions, and segmentation difficulty, the two final experiments were evaluated separately rather than as a direct comparison between farms.

5.7.3 DINOv3 and MegaDescriptor Video Comparison

A quantitative video comparison between DINOv3 and MegaDescriptor was conducted for Farm B. Both runs used the same 30-minute inference video, the same 191 cow identities and source gallery images, the same detector, custom YOLO11m segmentation, BoT-SORT settings, crop and mask-quality rules, matching thresholds, track-level settings, and RFID-based evaluation procedure.

Each embedding model created its own gallery prototypes from the same source gallery images. MegaDescriptor query embeddings were compared only with MegaDescriptor gallery prototypes, while DINOv3 query embeddings were compared only with DINOv3 gallery prototypes. This prevented embeddings from different feature spaces from being mixed. The only intended difference between the two runs was the Re-ID embedding model and its corresponding trained checkpoint.

5.7.4 Segmentation Comparisons for Farm B

The segmentation evaluation contained two parts. First, default YOLO11s, direct SAM3, and custom YOLO11m were compared qualitatively using the manual inspection criteria described in Section 5.4.3. Default YOLO11s was not included in a complete quantitative

video Re-ID run. Therefore, the thesis does not claim a quantitative Re-ID improvement over default YOLO11s.

Second, a controlled quantitative video Re-ID comparison was conducted between direct SAM3 and custom YOLO11m. The downstream Re-ID configuration was kept consistent across the two experiments. Both experiments used the same Farm B video, the same 191 cow identities and source gallery images, the same Farm B DINOv3 checkpoint, detector, BoT-SORT configuration, crop and mask-quality criteria, Re-ID thresholds, track-level rules, and output-evaluation procedure. The gallery prototypes were rebuilt from the segmented gallery crops produced by the corresponding segmentation method. Therefore, the primary experimental difference was the segmentation approach used to generate the gallery and video cow crops.

Direct SAM3 used the text prompt "cow", a confidence threshold of 0.30, and an inference image size of 1036 pixels. Its masks were thresholded at 0.40, dilated once with a 3×3 kernel, and valid mask regions were merged before the shared downstream Re-ID quality checks were applied. In the custom YOLO11m condition, the corresponding segmented crops were generated using the trained Farm B YOLO11m segmentation checkpoint.

5.7.5 Cross-Farm Transfer Experiment

The cross-farm experiment had two purposes. First, it evaluated whether the Farm A-trained DINOv3 model could be applied to Farm B without being retrained on Farm B cow identities. Second, it evaluated whether Farm B cow identities that were unseen during training of the Farm A DINOv3 model could be enrolled through their gallery images and then identified from Farm B video. In the standard Farm B experiment, the inference-gallery identities were already represented during training of the Farm B DINOv3 model. Therefore, this second aspect was evaluated only in the cross-farm experiment.

The Farm A-trained DINOv3 checkpoint was used as the embedding model for the Farm B gallery and Farm B inference video. The Farm B detector, custom YOLO11m segmentation, BoT-SORT configuration, thresholds, and evaluation procedure remained unchanged.

The Farm B gallery prototypes were rebuilt from the Farm B gallery images using the Farm A-trained DINOv3 checkpoint. The Farm B video query embeddings were generated using the same Farm A checkpoint. This ensured that gallery and query embeddings belonged to the same feature space. The result was compared with the standard Farm B run using the Farm B-trained DINOv3 checkpoint. Since only the Farm A-to-Farm B direction was tested, the experiment represents cross-farm transfer in one direction rather than full bidirectional cross-farm generalisation.

5.7.6 Video Ground Truth and Evaluation Procedure

The video-level evaluation was based on the biological cow identities displayed by the Re-ID system. Temporary BoT-SORT track IDs were used only for tracking and were not treated as final Cow Re-ID results.

For Farm A, the expected cow ID was included in each video filename. For example, a filename containing the label `ID430` indicated that cow 430 was the target cow in that video. After inference, the displayed cow ID was first compared with the filename label. The query cow visible in the video was also checked manually against the corresponding gallery images.

For Farm A, identification performance was evaluated at the target-video level. Each of the 15 enrolled cows was evaluated in one fixed one-minute daytime video and one fixed one-minute night-time video, resulting in 30 target-video trials. A target-video trial was considered successful when the displayed cow ID matched the expected cow ID and the manual visual check supported the match. Identification rates were calculated separately for the 15 daytime trials, the 15 night-time trials, and all 30 target-video trials. In addition, cross-condition recovery was evaluated by recording how many cow identities were successfully recovered in both conditions and in at least one of the two conditions. The identification-rate metric is defined in Section 2.10, and the video-level results for Farm A are presented in Section 6.4.

For Farm B, the RFID timestamp was expected to occur before the corresponding video timestamp because the cow first passed the RFID reader and then entered the camera view. A displayed cow identity was treated as temporally valid when the same cow ID appeared in the RFID record between 15 and 60 seconds before the video timestamp. The matching condition was therefore defined as

$$15 \leq t_{\text{video}} - t_{\text{RFID}} \leq 60 \text{ s}. \tag{5.1}$$

Here, t_{RFID} denotes the RFID detection time and t_{video} denotes the timestamp at which the cow identity was displayed by the Re-ID system.

For the Farm B gallery-identity evaluation, the 191 gallery identities were divided according to their presence in the RFID records of the evaluation video. Of these, 122 gallery identities were present and 69 were absent. The four evaluation cases were defined as follows:

- **True Positive (TP):** the gallery cow was present according to the RFID records and was recovered by the Re-ID system within the valid RFID time window.

- **False Negative (FN):** the gallery cow was present according to the RFID records but was not successfully recovered by the Re-ID system within the valid RFID time window.
- **False Positive (FP):** the gallery cow was absent according to the RFID records but its ID was displayed by the Re-ID system.
- **True Negative (TN):** the gallery cow was absent according to the RFID records and its ID was not displayed by the Re-ID system.

Precision, Recall, F1-score, Specificity, and Accuracy were then calculated from these values using the equations defined in Section 2.10.

The RFID records also contained 54 cows that were not represented in the 191-cow inference gallery. The correct Re-ID behaviour for these cows would be an Unknown decision. However, because the RFID events could not be linked directly to individual video tracks, an exact per-track open-set rejection rate was not calculated for these non-gallery cows. The Farm B RFID-supported video-level results are presented in Section 6.5.

5.8 Summary

This chapter described the implementation and experimental setup of the farm-specific Cow Re-ID system. A common PyTorch training pipeline was used for ResNet-50, MegaDescriptor, and DINOv3. Farm B pseudo-labels were generated with SAM3 and used to train the custom YOLO11m segmentation model. Separate inference scripts implemented the Farm A and Farm B gallery construction, tracking, segmentation, prototype matching, identity locking, and output procedures. The chapter also defined the image-level model comparison, final farm-specific video experiments, Farm B embedding-model comparison, segmentation comparisons, cross-farm transfer experiment, and ground-truth verification procedure. Chapter 6 presents the resulting measurements and their interpretation.

6 Experimental Evaluation and Discussion

This chapter presents and discusses the experimental results of the proposed farm-specific Cow Re-ID pipeline. The evaluation is divided into image-level and video-level experiments. The image-level experiments compare ResNet-50, MegaDescriptor, and DINOv3 on the two farm-specific datasets. The video-level experiments then evaluate the complete Re-ID pipeline using the experimental setup described in Section 5.7. Farm A mainly evaluates unseen-identity enrolment and performance across daytime and night-time recordings, while Farm B provides an RFID-supported evaluation of known gallery identities under several pipeline configurations.

The following sections first present the image-level Re-ID results in Section 6.2. The qualitative Farm B segmentation comparison is presented in Section 6.3. The final Farm A video results are analysed in Section 6.4, followed by the RFID-supported Farm B video evaluation in Section 6.5. Section 6.6 provides illustrative examples of the final video-inference output. Finally, Section 6.7 discusses the main findings across the experiments.

6.1 Evaluation Overview

The image-level evaluation uses two types of measures: classification metrics and embedding-based retrieval metrics. First, classification accuracy, macro precision, macro recall, and macro F1-score are calculated from the classification output on the held-out test images. The macro measures are calculated per cow identity and then averaged, so that each cow contributes equally to the final value. Second, Rank-1, Rank-5, and mAP evaluate the 256-dimensional embedding representation using query–gallery retrieval. These retrieval measures are conceptually closer to the final inference procedure because the video pipeline also identifies cows by comparing L2-normalised embeddings with gallery prototypes using cosine similarity. The definitions and equations for the evaluation metrics used in this chapter are provided in Section 2.10.

The image-level evaluation is closed-set because the same cow identities are represented in the training, validation, and test subsets, although the images themselves are different. The final video experiments are different. They use separate inference galleries and video

recordings and also contain non-gallery cows. Therefore, the video experiments provide a more practical evaluation of the complete Re-ID pipeline.

The Farm A and Farm B video results are not directly compared because the two farms differ in their datasets, inference galleries, ground-truth procedures, and environmental conditions. Together, the results show how the proposed system performs under different practical conditions.

6.2 Image-Level Re-ID Results

ResNet-50, MegaDescriptor, and DINOv3 were evaluated separately on Farm A and Farm B using the same image-level evaluation procedure described in Section 5.7.1. The classification measures evaluate the classifier output, while Rank-1, Rank-5, and mAP evaluate the learned embedding representation.

6.2.1 Farm A Image-Level Results

Table 6.1 presents the Farm A image-level results. DINOv3 achieved the highest test classification accuracy of 96.54%, the highest macro Recall of 94.49%, and the highest macro F1-score of 94.11%. ResNet-50 obtained a slightly higher macro Precision of 94.96% compared with 94.88% for DINOv3, but the difference was only 0.08 percentage points.

For embedding-based retrieval, DINOv3 also achieved the strongest result, with Rank-1 accuracy of 93.43%, Rank-5 accuracy of 97.73%, and mAP of 93.57%. ResNet-50 and MegaDescriptor achieved closely comparable results, while both were lower than DINOv3 in Rank-1 and mAP.

Table 6.1: Image-level Re-ID results for Farm A.

Model	Acc.	Macro P	Macro R	Macro F1	Rank-1	Rank-5	mAP
ResNet-50	96.10%	**94.96%**	93.83%	93.68%	92.17%	96.72%	92.56%
MegaDescriptor	96.10%	94.65%	93.75%	93.50%	91.92%	97.47%	92.62%
DINOv3	**96.54%**	94.88%	**94.49%**	**94.11%**	**93.43%**	**97.73%**	**93.57%**

The difference between the overall classification accuracy and macro F1-score is also informative. For DINOv3, classification accuracy was 96.54%, while macro F1-score was 94.11%. Since macro F1 gives equal importance to every cow identity, errors on cows with only a small number of test images can have a stronger effect on this measure than on overall accuracy. The result therefore shows that the model performed strongly overall, while performance was not perfectly uniform across all 216 identities.

6.2.2 Farm B Image-Level Results

The Farm B image-level results are shown in Table 6.2. DINOv3 achieved the highest classification accuracy of 99.09%, macro Precision and macro Recall of 98.69%, and macro F1-score of 98.46%. ResNet-50 also performed strongly, while MegaDescriptor achieved lower classification measures than the other two models.

For retrieval, both DINOv3 and MegaDescriptor achieved Rank-1 accuracy of 99.60%, Rank-5 accuracy of 100.00%, and mAP of 99.80%. ResNet-50 was only slightly lower, with Rank-1 accuracy of 99.19% and mAP of 99.44%.

Table 6.2: Image-level Re-ID results for Farm B.

Model	Acc.	Macro P	Macro R	Macro F1	Rank-1	Rank-5	mAP
ResNet-50	98.91%	98.56%	98.43%	98.24%	99.19%	99.60%	99.44%
MegaDescriptor	98.00%	97.39%	97.12%	96.78%	**99.60%**	**100.00%**	**99.80%**
DINOv3	**99.09%**	**98.69%**	**98.69%**	**98.46%**	**99.60%**	**100.00%**	**99.80%**

These values show that all three models can separate the Farm B identities very well within the image-level evaluation setting. However, the Farm B result must be interpreted in the context of the dataset construction. The Farm B Re-ID images were extracted from one continuous 30-minute recording. The training, validation, and test images were different, but they still came from the same recording environment and therefore contained less variation in illumination and acquisition conditions than the Farm A dataset.

For this reason, the near-perfect Farm B image-level scores should mainly be interpreted as strong discrimination within this recording environment. They do not by themselves demonstrate the same level of robustness under a separate continuous video. This point is important because the Farm B video-level results in Section 6.5 are clearly lower than the image-level results.

6.2.3 Interpretation of the Image-Level Comparison

Although the differences between the models were relatively small, DINOv3 showed the most consistent overall performance across classification and retrieval metrics in both farm datasets. It was therefore selected as the embedding model for the final video-based pipeline. MegaDescriptor was additionally evaluated at video level because it is specifically designed for animal re-identification and achieved the same Farm B Rank-1, Rank-5, and mAP values as DINOv3, as shown in Table 6.2. This made it a strong alternative for examining whether similar image-level retrieval performance would also be maintained under video-based

inference. ResNet-50 was retained as the conventional CNN baseline in the image-level comparison.

The two farms nevertheless represent different levels of image diversity. Farm A contained images collected from different recordings and under different daytime and night-time conditions. Its image-level results therefore reflect a more varied appearance setting. In contrast, Farm B had less recording diversity because its images were obtained from a single continuous recording under a more homogeneous acquisition condition. The available Re-ID image data came from one 30-minute recording, while a second 30-minute recording was reserved for the final video inference. Using the first recording was therefore necessary to construct the Farm B Re-ID dataset, but its limited visual diversity should be considered when interpreting the image-level performance.

The distinction between the classifier and the embedding representation is also important. The classification layer is not used to identify cows during video inference. Instead, the 256-dimensional embedding is compared with gallery prototypes using cosine similarity. However, the embedding is learned during supervised model training, so limited training diversity can still influence the representation. The very high Farm B Rank-1, Rank-5, and mAP values therefore show that the embedding works very well within the image-level recording environment, but they do not remove the need for a separate video-level evaluation.

6.3 Farm B Segmentation Evaluation

The Farm B segmentation evaluation was qualitative because no manually annotated pixel-level ground-truth masks were available. The comparison therefore focuses on whether the visible cow body was retained, whether important body regions were lost, whether background or vertical bars remained in the crop, and whether the resulting segmented image was suitable for Re-ID embedding extraction. The examples shown in this section were selected as representative cases of the behaviour observed during manual inspection.

6.3.1 Default YOLO11s and Custom YOLO11m

Figure 6.1 shows one representative Farm B example processed by the default YOLO11s segmentation model and the custom YOLO11m model. The source image contains strong vertical metal-bar occlusion. The default YOLO11s result retains only part of the visible cow body, whereas the custom YOLO11m result preserves a larger and more complete set of visible body regions between the bars.



(a) Source image.



(b) Default YOLO11s.



(c) Custom YOLO11m.

Figure 6.1: Representative qualitative comparison between default YOLO11s and the custom YOLO11m segmentation model for Farm B.

This visual comparison supports the use of the farm-specific segmentation model under the strong metal-bar occlusion. However, the figure is a qualitative example and is not used to claim a numerical segmentation improvement over the default YOLO11s model.

6.3.2 Direct SAM3 and Custom YOLO11m

Figure 6.2 presents two representative cows, IDs 9440 and 9868, processed by direct SAM3 and the custom YOLO11m model. In both examples, direct SAM3 returns a usable visible-cow mask but loses more visible body regions and produces a more fragmented result. The custom YOLO11m result retains a more complete set of visible cow-body regions across the bar occlusions.

The qualitative result favours the custom YOLO11m segmentation, but the final importance of this difference must be judged through the downstream Re-ID experiment. As shown later in Section 6.5.2, direct SAM3 produced video Re-ID performance close to the custom YOLO11m pipeline. This suggests that a visually more complete segmentation does not necessarily produce a proportionally large improvement in final identity recovery.



(a) Cow 9440: source. (b) Cow 9440: Direct SAM3. (c) Cow 9440: Custom YOLO11m.



(d) Cow 9868: source. (e) Cow 9868: Direct SAM3. (f) Cow 9868: Custom YOLO11m.

Figure 6.2: Representative qualitative comparison between direct SAM3 and custom YOLO11m for Farm B under vertical metal-bar occlusion.

6.4 Farm A Video Re-ID Evaluation

The final Farm A video experiment evaluates whether the DINOv3-based pipeline can re-identify cows whose identities were not used during model training, validation, or image-level testing. Fifteen such unseen cow identities were enrolled in the inference gallery. Each cow was evaluated using two separate fixed one-minute videos: one daytime video and one night-time video. The final evaluation therefore contained 30 target-video trials.

A target-video trial was considered successful when the displayed cow ID matched the expected cow ID and the manual visual check supported the match. Each one-minute video was associated with a specific target cow, whose ID was indicated in the video filename and manually verified against the corresponding gallery images. Other cows could also appear in the same video, but they were not considered the target identity for that trial. Since synchronised RFID or other ground-truth identity information was not available for all cows appearing in the Farm A videos, the evaluation focused on whether the expected target cow was successfully re-identified. Therefore, TP, TN, FP, and FN values were not calculated for Farm A.

6.4.1 Daytime and Night-time Results

Table 6.3 presents the outcome for all 15 cows. Eleven cows were identified in the daytime videos and twelve were identified in the night-time videos.

Table 6.3: Farm A cow-level identification results under daytime and night-time conditions.

Cow ID	Daytime	Night-time
420	Identified	Identified
425	Identified	Identified
430	Identified	Identified
433	Identified	Identified
436	Not identified	Not identified
446	Identified	Not identified
453	Identified	Identified
465	Identified	Identified
562	Identified	Identified
568	Not identified	Identified
593	Identified	Identified
596	Not identified	Identified
604	Identified	Identified
605	Not identified	Not identified
625	Identified	Identified

The daytime identification rate was

$$IR_{\text{day}} = \frac{11}{15} \times 100 = 73.33\%. \tag{6.1}$$

The night-time identification rate was

$$IR_{\text{night}} = \frac{12}{15} \times 100 = 80.00\%. \tag{6.2}$$

Across all 30 target-video trials, 23 were successfully identified:

$$IR_{\text{overall}} = \frac{23}{30} \times 100 = 76.67\%. \tag{6.3}$$

Table 6.4 summarises the daytime, night-time, and overall identification results. It also reports the corresponding identification rates for each condition.

Table 6.4: Summary of Farm A video-level identification performance.

Condition	Identified	Not identified	Total	Identification rate
Daytime	11	4	15	73.33%
Night-time	12	3	15	80.00%
Overall	23	7	30	76.67%

The night-time identification rate was 6.67 percentage points higher than the daytime identification rate. However, this difference should not be interpreted as evidence that night-time conditions generally improve Cow Re-ID performance, since only 15 cow identities were evaluated in each condition. Instead, the results show that the pipeline did not exhibit a systematic loss of performance under the evaluated night-time recordings.

6.4.2 Cross-Condition Identity Recovery

Ten of the 15 cows were successfully identified in both daytime and night-time videos. The both-condition recovery rate was therefore

$$R_{\text{both}} = \frac{10}{15} \times 100 = 66.67\%. \tag{6.4}$$

Cow 446 was identified only during daytime, while cows 568 and 596 were identified only during night-time. Cows 436 and 605 were not identified in either condition. When recovery in at least one of the two conditions is considered, 13 of the 15 unseen enrolled cows were successfully recovered:

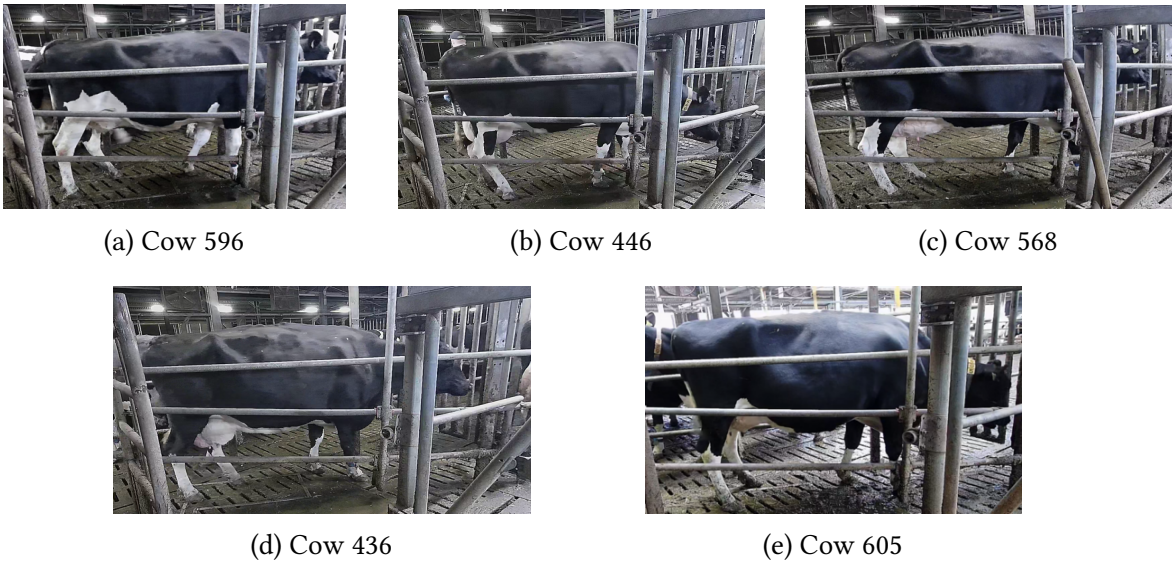
$$R_{\text{at least once}} = \frac{13}{15} \times 100 = 86.67\%. \tag{6.5}$$

The cross-condition result provides additional information beyond the overall 76.67% target-video identification rate. Two-thirds of the cows were consistently recovered under both evaluated illumination conditions, while 86.67% were recovered in at least one condition. This supports the ability of the gallery-based pipeline to enrol and identify cow identities that were not part of model training.

6.4.3 Analysis of Difficult Farm A Identities

The five cows that were not consistently recovered were manually inspected: IDs 436, 446, 568, 596, and 605. As shown in Figure 6.3, a common visual characteristic was that their visible side-body region was predominantly black, with relatively few distinctive coat-pattern

regions from the camera viewpoint. Cows 446, 568, and 596, which were identified in at least one of the two conditions, showed some additional contrasting coat regions, particularly around the lower body and legs. In comparison, cows 436 and 605, which were not identified in either condition, showed particularly limited distinctive side-body coat information.



(a) Cow 596 (b) Cow 446 (c) Cow 568

(d) Cow 436 (e) Cow 605

Figure 6.3: Difficult Farm A identities. The first-row cows were identified in at least one condition, whereas the second-row cows were not identified in either condition.

These qualitative observations are relevant because the Re-ID decision is based on visual features represented in the embedding. Previous studies have shown that distinctive coat patterns can provide useful identity information for Holstein-Friesian cattle [3, 4]. In the present experiment, this was also observed in the cows that were difficult to identify consistently, which had relatively few distinctive side-body coat features from the available camera viewpoint. A largely uniform side-body appearance may therefore make an identity more difficult to distinguish in the embedding space. Changes in illumination may further affect the visibility of the limited distinctive features that are available.

The observed coat appearance provides a plausible explanation for the inconsistent identification of these cows, but it does not establish limited distinctive coat information as the only cause. Pose, the amount of visible body surface, motion blur, segmentation quality, tracking duration, and differences between gallery and query observations may also affect the final identity decision.

6.4.4 Interpretation and Limitations of the Farm A Evaluation

The Farm A experiment provides two main findings about the proposed pipeline. First, the 15 cow identities enrolled in the inference gallery were not used during DINOv3 training, validation, or image-level testing. The successful recovery of 13 of these 15 cows in at least one of the evaluated videos shows that new cow identities can be enrolled through the inference gallery without retraining the model. Second, ten cows were successfully recovered in both daytime and night-time videos, indicating that the pipeline remained effective under both evaluated illumination conditions.

A limitation of the Farm A evaluation is that ground-truth identity information was not available for all cows appearing in the videos. The expected target cow for each one-minute video was known, but other cows could also appear in the same video without verified identity information. Therefore, a conventional TP, TN, FP, and FN analysis could not be reliably calculated for all cows in the Farm A videos. In addition, only one daytime and one night-time target video were evaluated for each cow. The daytime and night-time results should therefore be interpreted as performance under the evaluated recordings rather than as proof of robustness under all possible illumination conditions.

6.5 Farm B RFID-Supported Video Re-ID Evaluation

Farm B provides the main RFID-supported quantitative evaluation of the complete video pipeline. The inference gallery contained 191 cow identities. During the 30-minute evaluation video, 122 of these gallery identities appeared in the RFID records, while 69 did not. In addition, the RFID records contained 54 cows that were not included in the gallery.

A Re-ID identity recovery was considered temporally valid when the same cow ID was displayed between 15 and 60 seconds after the RFID event. The detailed ground-truth procedure is described in Section 5.7.6. The evaluation therefore considers whether cow identities included in the gallery were recovered within the allowed RFID time window, rather than evaluating each tracked cow separately.

6.5.1 Overall Farm B Results

Table 6.5 compares the four evaluated Farm B configurations: the final DINOv3 pipeline with custom YOLO11m segmentation, the same DINOv3 pipeline with direct SAM3 segmentation, the cross-farm DINOv3 configuration, and the MegaDescriptor-based configuration with custom YOLO11m segmentation.

Table 6.5: RFID-supported temporal identity-recovery results for Farm B.

Configuration	TP	TN	FP	FN	Precision	Recall	F1	Specificity	Accuracy
Custom YOLO11m	87	63	6	35	93.55%	**71.31%**	**80.93%**	91.30%	**78.53%**
Direct SAM3	84	64	5	38	94.38%	68.85%	79.62%	92.75%	77.49%
Cross-Farm DINOv3	75	66	3	47	**96.15%**	61.48%	75.00%	**95.65%**	73.82%
MegaDescriptor	63	59	10	59	86.30%	51.64%	64.62%	85.51%	63.87%

The final custom YOLO11m configuration achieved the highest Recall of 71.31%, the highest F1-score of 80.93%, and the highest Accuracy of 78.53%. It successfully recovered 87 of the 122 RFID-confirmed gallery identities. Its Precision was also high at 93.55%, indicating that the higher number of recovered identities did not result in many false-positive gallery identities under the RFID evaluation.

Direct SAM3 produced results close to the custom segmentation pipeline. Cross-Farm DINOv3 produced the highest Precision and Specificity but lower Recall. MegaDescriptor produced the lowest Recall, F1-score, and Accuracy among the four configurations.

6.5.2 Quantitative Effect of the Segmentation Method

As described in Section 5.7.4, the downstream Re-ID configuration was kept consistent across the custom YOLO11m and direct SAM3 experiments, while the gallery and video cow crops were generated using the corresponding segmentation method. Custom YOLO11m recovered 87 identities, while direct SAM3 recovered 84. The F1-scores were 80.93% and 79.62%, respectively, and the Accuracy values were 78.53% and 77.49%. Therefore, the difference in Accuracy was 1.04 percentage points.

This result is important when interpreted together with the qualitative examples in Section 6.3. Custom YOLO11m generally produced more complete visible-cow masks, whereas direct SAM3 sometimes produced more fragmented masks. However, the difference in the final Re-ID performance was modest. This suggests that direct SAM3 often retained enough discriminative cow-body information for the DINOv3 embedding model to recover the correct identity. In addition, the video pipeline compares query embeddings from multiple accepted frames with the gallery prototypes and accumulates the resulting similarity evidence over time, as described in Section 4.5.3. This may reduce the influence of a single less informative observation.

Nevertheless, the custom YOLO11m pipeline recovered three more RFID-confirmed gallery identities and achieved higher Recall and F1-score. The qualitative inspection, together with

the slightly better Re-ID performance, supported the selection of custom YOLO11m for the final Farm B configuration.

6.5.3 DINOv3 and MegaDescriptor Video Comparison

The comparison between DINOv3 and MegaDescriptor used the same Farm B video, gallery source images, custom YOLO11m segmentation, tracking settings, and evaluation procedure. Each embedding model created and used its own gallery prototypes.

DINOv3 with custom YOLO11m recovered 87 of the 122 RFID-confirmed gallery identities, while MegaDescriptor recovered 63. Recall decreased from 71.31% to 51.64%, a difference of 19.67 percentage points. MegaDescriptor also produced ten false-positive gallery identities compared with six for DINOv3. Its F1-score was 64.62%, compared with 80.93% for DINOv3.

The result shows that MegaDescriptor was less effective than the farm-specific DINOv3 model under the evaluated Farm B video conditions. The experiment does not identify a single reason for this difference, but it shows that the choice of embedding model had an important effect on the overall video Re-ID performance.

6.5.4 Cross-Farm Transfer from Farm A to Farm B

The cross-farm experiment had two related purposes. First, it evaluated whether the DINOv3 model trained on Farm A could transfer to Farm B without being retrained on Farm B cow identities. Second, it evaluated whether Farm B cow identities that had never been used during training of the Farm A DINOv3 model could be enrolled using their gallery images and then identified from Farm B video. This second aspect was not evaluated in the standard Farm B experiment presented in Section 6.5.1, because all identities used in the Farm B inference gallery were already represented during training of the Farm B DINOv3 model. In addition, the final Farm B inference gallery was constructed from the same source image dataset used for Farm B model development.

For the cross-farm experiment, the Farm B gallery prototypes were created using the Farm A-trained DINOv3 model, and the same model was used to extract the Farm B video query embeddings. This ensured that the gallery and video embeddings were generated in the same feature space. All other Farm B inference components remained unchanged.

The cross-farm configuration recovered 75 of the 122 RFID-confirmed gallery identities, corresponding to a Recall of 61.48%. This was lower than the 71.31% Recall of the standard Farm B-trained DINOv3 configuration. However, the cross-farm model produced only three

false-positive gallery identities and therefore achieved the highest Precision of 96.15% and the highest Specificity of 95.65% among the evaluated Farm B configurations.

These higher values were accompanied by lower Recall, F1-score, and Accuracy. Therefore, the cross-farm configuration should not be interpreted as performing better overall than the Farm B-trained model.

The results provide evidence for both aspects of the cross-farm experiment. The Farm A-trained DINOv3 model recovered 75 Farm B identities even though the Farm B cow identities had not been used during its Re-ID training. This provides evidence that cross-farm transfer was possible and that Farm B identities unseen during training of the Farm A DINOv3 model could be enrolled through the inference gallery without retraining the DINOv3 model. However, the Farm A-trained model recovered fewer identities than the Farm B-trained DINOv3 model. Since only the Farm A-to-Farm B direction was evaluated, this result is interpreted as one-directional cross-farm transfer rather than full cross-farm generalisation.

6.5.5 Limitations of the RFID-Supported Evaluation

The RFID-supported Farm B evaluation provides a consistent external temporal reference, but it does not give an exact identity label for every video track. A prediction that falls within the valid RFID time window is consistent with the RFID evidence, but the available records do not establish a direct one-to-one correspondence between each RFID event and a specific tracked cow in the video.

This limitation is especially important for the 54 RFID-confirmed non-gallery cows. The correct Re-ID behaviour for these cows would be an Unknown decision. However, an exact open-set rejection rate could not be calculated because the RFID events for these cows could not be linked to specific tracked cows in the video. Therefore, the reported TP, TN, FP, FN, Precision, Recall, F1-score, Specificity, and Accuracy values should be interpreted according to the RFID-based gallery-identity evaluation procedure rather than as conventional per-track multiclass metrics.

6.6 Illustrative Video Output Examples

Figure 6.4 shows illustrative screenshots from the final video-inference output for Farm A and Farm B. The examples illustrate how the proposed pipeline presents the identified cow during video processing. In these examples, the output frame displays the assigned cow ID together with the corresponding similarity score and the tracked cow region.

The Farm A example shows cow ID 430 identified in one of the final inference videos, while the Farm B example shows cow ID 8886 identified under the stronger vertical metal-bar occlusion conditions of Farm B. These screenshots are included as qualitative examples of the final system output and are not used as an additional performance measure. The quantitative video-level results are presented in the preceding Farm A and Farm B evaluation sections.



(a) Farm A example: cow ID 430.



(b) Farm B example: cow ID 8886.

Figure 6.4: Illustrative screenshots from the final video-based Cow Re-ID output for Farm A and Farm B. The displayed annotations show the assigned cow identity, similarity score, and tracked cow region.

6.7 Discussion

The experiments show that image-level and video-level Re-ID performance should be interpreted separately. At image level, DINOv3 achieved strong classification and retrieval results on both farms. However, the final video pipeline introduced additional difficulties, including detection and segmentation quality, occlusion, tracking duration, changes in appearance, open-set observations, and the need to accumulate stable evidence across frames.

6.7.1 Effect of Dataset Diversity and Recording Conditions

Farm A and Farm B differed substantially in the diversity of the available Re-ID image data. Farm A contained images collected from different recordings, dates, and daytime and night-time conditions. Farm B had only two available 30-minute recordings: one was used to construct the Re-ID image dataset and inference gallery, while the second independent recording was reserved for video inference. As a result, the Farm B image dataset contained less variation in recording conditions.

This difference should be considered when interpreting the near-perfect Farm B image-level results, as they do not directly represent real-world video Re-ID performance. The very high

Farm B classification and retrieval scores indicate that the model could distinguish the cow identities very well within the image-level dataset. In the separate video experiment, however, the final DINOv3 pipeline recovered 87 of 122 RFID-confirmed gallery identities. The lower video-level result indicates that strong closed-set image retrieval does not automatically translate into the same level of performance in continuous video.

The limited Farm B image diversity may have contributed to this gap, but the experiment does not isolate dataset diversity as the only cause. Farm B also contains stronger metal-bar occlusion and introduces the additional difficulties of continuous tracking, segmentation quality, and the presence of cows that are not included in the inference gallery. Therefore, dataset diversity is treated as a plausible contributing factor rather than a proven single cause.

Farm A provides complementary evidence. The 15 cow identities enrolled in the Farm A inference gallery were unseen during training, validation, and image-level testing, and the same cow identities were evaluated in separate daytime and night-time videos. Ten cows were recovered in both conditions and 13 were recovered in at least one condition. This result is consistent with the greater appearance diversity represented in the Farm A dataset and suggests that more varied recording conditions may be useful for learning a more robust appearance representation.

6.7.2 Importance of Visual Coat Information

The Farm A error analysis also suggests an important limitation of appearance-based Cow Re-ID. Cows with predominantly uniform black side-body regions provided fewer distinctive coat-pattern cues from the available camera viewpoint. These identities were more difficult to recover consistently. These observations suggest that the ability of the embedding to distinguish identities depends partly on how much identity-specific visual information is visible in the segmented body region.

Previous studies on Holstein-Friesian cattle have shown that distinctive coat patterns can provide useful visual information for individual identification [3, 4]. The Farm A observations suggest a similar practical limitation: when few distinctive visual features are visible from the available camera viewpoint, appearance-based Re-ID may become more difficult. Additional viewpoints, longer observations, or other complementary identity information may be useful in future work.

6.7.3 Segmentation Quality and Re-ID Performance

The Farm B segmentation experiments show that visual mask quality and final Re-ID performance are related, but not in a simple one-to-one manner. Custom YOLO11m generally produced more complete visible-cow masks than direct SAM3 in the selected qualitative examples. Nevertheless, the difference between the two complete Re-ID pipelines was small: 78.53% versus 77.49% Accuracy and 80.93% versus 79.62% F1-score.

In the evaluated Farm B experiment, the results suggest that less complete segmentation did not necessarily prevent successful identity recovery. If enough discriminative body information remained, the query embedding could still support correct gallery matching. Track-level evidence accumulation may also reduce the influence of a single less informative observation. At the same time, the custom YOLO11m pipeline recovered the largest number of enrolled Farm B identities and was therefore retained for the final Farm B configuration.

6.7.4 Overall Interpretation

Taken together, the results support the use of DINOv3 as the main embedding model in the proposed pipeline. The Farm A evaluation shows that the gallery-based pipeline can enrol cow identities that were not used during model training and recover most of them across the evaluated daytime and night-time recordings. The Farm B evaluation shows that the complete system can recover a substantial proportion of RFID-confirmed gallery identities under strong metal-bar occlusion, while also showing that continuous-video performance is more challenging than closed-set image-level retrieval.

The results indicate that final Re-ID performance can be influenced by several parts of the pipeline. These include the embedding model, segmentation quality, available visual coat information, dataset diversity, and temporal evidence across the track. This is an important practical finding because strong image-level performance does not always lead to equally strong video-level performance.

The numerical results of this thesis should not be compared directly with those reported in previous cattle Re-ID studies because the datasets, experimental settings, and evaluation protocols are different. Therefore, the results are interpreted within the experimental setup of this thesis rather than as a direct performance comparison with previous work.

The results also address the three research questions introduced in Chapter 1. For RQ1, DINOv3 showed the strongest overall image-level performance across the two farm datasets. For RQ2, the final pipeline correctly identified 23 of 30 Farm A target-video trials and recovered 87 of 122 RFID-confirmed Farm B gallery identities. For RQ3, custom YOLO11m

achieved slightly higher Farm B video Re-ID performance than direct SAM3. The quantitative video comparison did not include default YOLO11s; it was assessed qualitatively.

6.8 Summary

This chapter evaluated the proposed Cow Re-ID pipeline at both image and video levels. DINOv3 achieved the strongest overall image-level results across the two farm-specific datasets. The Farm A video experiment recovered 13 of 15 unseen enrolled cows in at least one illumination condition and 10 cows under both daytime and night-time conditions. The Farm B RFID-supported evaluation showed that the final DINOv3 and custom YOLO11m configuration recovered 87 of 122 RFID-confirmed gallery identities and achieved an F1-score of 80.93%. Direct SAM3 produced closely comparable final Re-ID results despite visually less complete masks in the representative qualitative examples. The cross-farm experiment showed that Farm B identities unseen during training of the Farm A DINOv3 model could be enrolled through the inference gallery and recovered from Farm B video, although fewer identities were recovered than with the Farm B-trained DINOv3 model. MegaDescriptor produced the lowest Farm B video performance among the evaluated embedding configurations.

The combined results indicate that practical Cow Re-ID performance can be influenced not only by image-level model performance, but also by data diversity, visible identity features, segmentation quality, and temporal evidence across video tracks. The final chapter summarises the contributions and limitations of the thesis and presents directions for future work.

7 Conclusion and Future Work

This thesis developed and evaluated an end-to-end computer-vision pipeline for individual cow re-identification in dairy farm environments. The main goal was to identify individual cows from side-view farm videos using their visual appearance. The proposed system combines cow localisation and segmentation, DINOv3 embedding extraction, gallery-based matching, BoT-SORT tracking, and track-level identity stabilisation.

The system was developed and evaluated using two farm-specific scenarios, referred to as Farm A and Farm B. The two farms contained different recording conditions and different levels of segmentation difficulty. Farm A included daytime and night-time recordings with less severe occlusion, while Farm B contained stronger vertical metal-bar occlusion. Therefore, the two farms were treated as separate farm-specific scenarios, with separate DINOv3 models and segmentation strategies.

7.1 Conclusion

For image-level evaluation, ResNet-50, MegaDescriptor, and DINOv3 were compared using classification and retrieval-based metrics. DINOv3 produced the strongest overall results in both farm datasets and was therefore selected as the main embedding model for the final video-inference pipeline. For Farm A, DINOv3 achieved a Rank-1 accuracy of 93.43% and an mAP of 93.57%. For Farm B, it achieved a Rank-1 accuracy of 99.60% and an mAP of 99.80%. However, the high Farm B image-level results were obtained from images extracted from the same recording environment and should therefore not be interpreted as equivalent performance in continuous farm video.

The Farm A video experiment evaluated whether cow identities that were not used during DINOv3 training, validation, or image-level testing could still be identified after being added to the inference gallery. Across 30 daytime and night-time target videos, the correct cow was identified in 23 videos, giving an overall identification rate of 76.67%. Thirteen of the 15 enrolled cows were identified in at least one condition, while ten cows were identified under both daytime and night-time conditions. These results show that the gallery-based approach can support the enrolment of new cow identities without retraining the Re-ID model.

The Farm A analysis also suggested that the amount of distinctive visual information may affect identification performance. The cows that were difficult to identify consistently had relatively few distinctive side-body coat features from the available camera viewpoint. This may have made these identities more difficult to distinguish. Changes in illumination,

visible body area, pose, segmentation quality, and other differences between gallery and video observations may also affect the final identity decision.

Farm B provided a continuous-video experiment under stronger metal-bar occlusion. A custom YOLO11m segmentation model trained using SAM3-generated pseudo-labels was used for cow-body segmentation in this environment. With the final DINOv3 and custom YOLO11m configuration, 87 of the 122 RFID-confirmed gallery identities were recovered. The system achieved a Precision of 93.55%, a Recall of 71.31%, an F1-score of 80.93%, and an Accuracy of 78.53%.

The Farm B segmentation comparison showed that visually more complete segmentation did not produce a proportionally large improvement in final Re-ID performance. Direct SAM3 recovered 84 RFID-confirmed gallery identities, compared with 87 using the custom YOLO11m model. This suggests that DINOv3 can still produce useful identity information when enough of the cow body remains visible in the segmented image. However, the custom YOLO11m configuration recovered more RFID-confirmed gallery identities and achieved higher Recall, F1-score, and Accuracy than direct SAM3. It was therefore retained in the final Farm B pipeline.

DINOv3 also performed better than MegaDescriptor during the Farm B video comparison. Under the same segmentation and inference conditions, DINOv3 recovered 87 RFID-confirmed gallery identities, while MegaDescriptor recovered 63. This supports the selection of DINOv3 as the main embedding model for the proposed system.

A cross-farm experiment was also performed by applying the Farm A-trained DINOv3 model to Farm B. The cross-farm configuration recovered 75 of the 122 RFID-confirmed gallery identities, compared with 87 for the Farm B-trained model. This shows that Farm B identities unseen during training of the Farm A DINOv3 model could still be enrolled through their gallery images and identified from Farm B video, indicating the possibility of cross-farm transfer. However, the Farm B-trained model achieved better identity recovery under the evaluated Farm B conditions.

Overall, the results indicate that final Cow Re-ID performance can be influenced by several parts of the complete pipeline. These include the embedding model, segmentation quality, visible coat information, tracking, dataset diversity, and temporal identity evidence. Strong image-level performance alone does not always lead to equally strong video-level performance.

The main contribution of this thesis is therefore the development and evaluation of a complete gallery-based video Cow Re-ID pipeline. The system uses cow-level gallery prototypes, allows uncertain identities to remain Unknown, and uses evidence from multiple video frames before displaying a cow identity.

7.2 Future Work

Several areas of the proposed Cow Re-ID system can be improved in future work. These improvements can focus on increasing dataset diversity, improving video-level evaluation, and making the system more robust under challenging farm conditions.

More diverse training and evaluation data should be collected. This is particularly important for Farm B, where the available Re-ID image dataset was created from a limited recording period. Images and videos collected over different days, lighting conditions, seasons, and camera settings would provide a stronger evaluation of model generalisation.

More detailed video-level ground truth would improve the evaluation of the complete pipeline. The RFID timestamps used for Farm B provided an external identity reference, but individual RFID events could not be linked directly to individual video tracks. Future work could use manual track-level annotations or improved synchronisation between RFID and video data. This would allow more accurate measurement of identity switches, false identifications, and Unknown decisions.

The Farm A analysis also suggested that cows with relatively few distinctive side-body coat features were more difficult to identify consistently. Future work could therefore use additional camera viewpoints or combine information from different visible body regions. Vision-based Re-ID could also be combined with RFID or other identity information when visual appearance alone is not sufficiently distinctive.

Tracking under heavy occlusion could also be improved. BoT-SORT provided temporary track IDs in the current pipeline, but cows can overlap, disappear behind farm structures, and later reappear. Future work could compare alternative tracking methods or investigate the use of DINOv3 embeddings for reconnecting separated track segments.

Finally, the cross-farm evaluation should be extended to additional farms and longer continuous recordings. This would help determine how well the learned cow representations generalise across different cameras, environments, and herds, and how much farm-specific training is required.

In conclusion, this thesis shows that a gallery-based Cow Re-ID pipeline can identify individual cows from farm videos and can also enrol new cow identities without retraining the Re-ID model. The experiments further show that strong image-level performance alone does not guarantee equally strong performance in continuous video, where segmentation, tracking, visible identity information, and temporal evidence also become important. Future improvements in dataset diversity, video ground truth, tracking, and cross-farm evaluation could make the system more reliable for practical dairy farm monitoring.

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I hereby declare that all AI-supported tools used during the preparation of this thesis are listed below. I reviewed, verified, and edited all AI-supported content before including it in the thesis. The scientific decisions, methodology, implementation, evaluation, and final responsibility for the thesis remain my own.

Overview of IT/AI Tools

Tool Name	Tool Version	Purpose
ChatGPT	GPT-5.5 Thinking and GPT-5.6 Sol	Assisted with language refinement, thesis structure, consistency checking, and the creation of schematic figures based on author-provided technical descriptions, as well as the analysis and debugging of Python and LaTeX code.

Declaration of Independent Work

I hereby declare that I prepared this thesis independently and without unauthorised assistance. I used only the sources and aids listed in this thesis. All passages taken directly or indirectly from published or unpublished sources are identified as such. I also declare that the submitted electronic version corresponds to the printed version.

Rostock, 19. August 2026
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